

Chapter 11 Part 4: Titration

1.

Initial buret reading	2.17 mL
Final buret reading	22.17 mL

An antacid tablet containing $\text{Mg}(\text{OH})_2(s)$ (molar mass 58.3 g/mol) is titrated with a 0.100 M solution of $\text{HNO}_3(aq)$. The end point is determined by using an indicator. Based on the data in the table above, what was the mass of the $\text{Mg}(\text{OH})_2(s)$ in the antacid tablet?

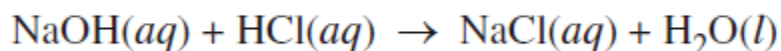
(A) 0.0583 g



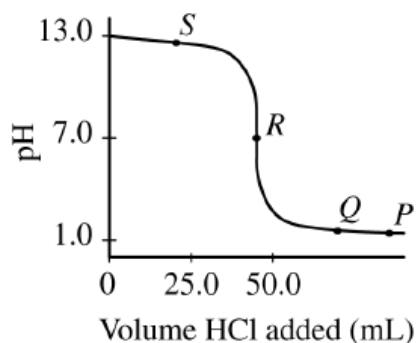
(B) 0.583 g

(C) 5.83 g

(D) 58.3 g



To determine the concentration of a $\text{NaOH}(aq)$ solution, a student titrated a 50. mL sample with 0.10 M $\text{HCl}(aq)$. The reaction is represented by the equation above. The titration is monitored using a pH meter, and the experimental results are plotted in the graph below.



2. A student mixes a 10.0 mL sample of 1.0 M $\text{NaOH}(aq)$ with a 10.0 mL sample of 1.0 M $\text{HCl}(aq)$ in a polystyrene container. The temperature of the solutions before mixing was 20.0°C. If the final temperature of the mixture is 26.0°C, what is the experimental value of ΔH_{rxn}° ? (Assume that the solution mixture has a specific heat of 4.2 J/(g•K) and a density of 1.0 g/mL.)

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(A) $-50. \text{ kJ/mol}_{\text{rxn}}$



(B) $-25 \text{ kJ/mol}_{\text{rxn}}$

(C) $-5.0 \times 10^4 \text{ kJ/mol}_{\text{rxn}}$

(D) $-5.0 \times 10^2 \text{ kJ/mol}_{\text{rxn}}$

3.

Trial	Volume of 0.10 M HCl	Volume of 0.10 M NaOH	Amount of Heat Released
1	50. mL	50. mL	X
2	100. mL	50. mL	Y

A student conducted an experiment to determine $\Delta H_{\text{rxn}}^{\circ}$ for the reaction between $\text{HCl}(aq)$ and $\text{NaOH}(aq)$.

The student ran two trials using the volumes of $\text{HCl}(aq)$ and $\text{NaOH}(aq)$ indicated in the table above, and determined the amount of heat released. Which of the following best explains the relationship between X and Y?

(A) $Y = 2X$, because the volume of $\text{HCl}(aq)$ used in trial 2 is twice the volume used in trial 1.

(B) $Y = X$, because the number of moles of acid and base reacting with each other is the same in both trials.



(C) $Y = 2X / 3$, because the heat is distributed over more particles in trial 2 than in trial 1.

(D) The relationship between X and Y cannot be predicted.

4. At the point labeled R on the pH curve, which of the following ions are present in the reaction mixture at a concentration greater than 0.01 M ?

(A) Na^+ and Cl^- only



(B) Na^+ , Cl^- and H^+ only

(C) Na^+ , Cl^- and OH^- only

(D) Na^+ , Cl^- , H^+ and OH^-

5. One student titrated the $\text{NaOH}(aq)$ with $1.0 \text{ M HCl}(aq)$ instead of $0.10 \text{ M HCl}(aq)$. How would the student's titration curve differ from the original curve?

(A) The initial pH would be 11 instead of 13.

(B) The pH at the equivalence point would be 5 instead of 7.

(C) The pH far beyond the equivalence point would be higher than in the original curve.

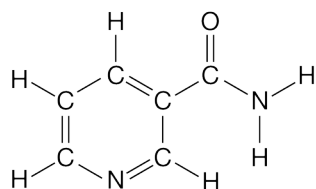
(D) The pH far beyond the equivalence point would be lower than in the original curve.



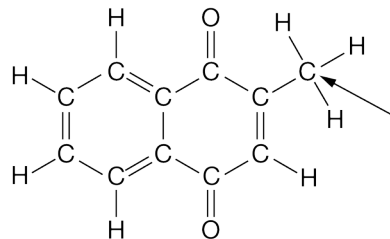
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6. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound W and Compound Y. Structures of the compounds are shown below.



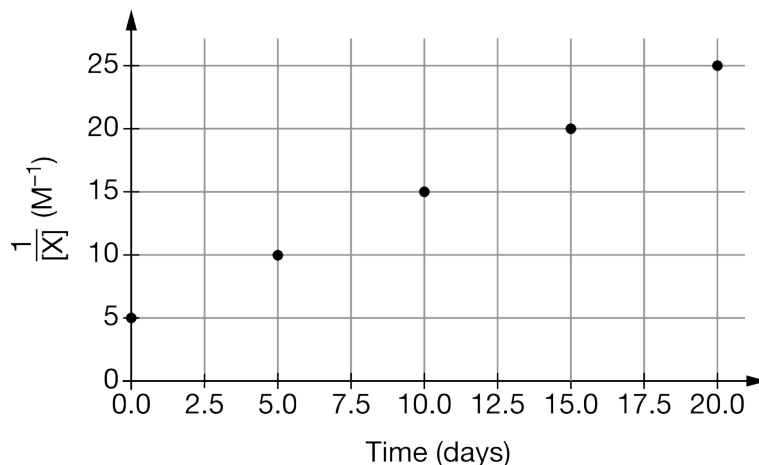
Compound W
(molar mass = 122 g/mol)



Compound Y
(molar mass = 172 g/mol)

- (a) Identify the hybridization of the valence orbitals of the carbon atom indicated with the arrow in the diagram of compound Y.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound W has a greater boiling point than compound Y. The student claims that the difference is because compound Y exhibits stronger hydrogen bonding than compound W. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

A different compound found in food, compound X, is known to decompose over time. To study the reaction, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting $1/[X]$ versus time.



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(d) The student proposes that the reaction is second order with respect to compound X. Do you agree or disagree? Justify your answer using the data in the graph above.

(e) Write the rate law expression for the decomposition of compound X that is consistent with the graph above.

(f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.

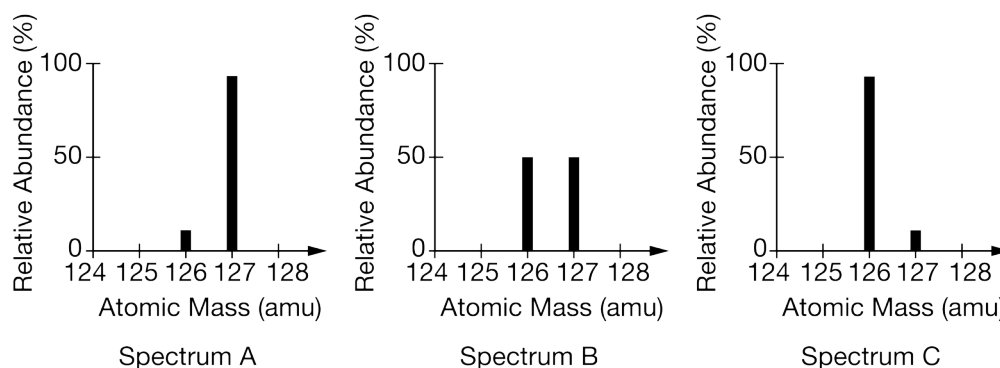
(g) The student claims that if the experiment were conducted again at a higher temperature, the value for the rate constant, k , would decrease. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the higher temperature conditions.

A student titrates a solution of compound X to determine the concentration of X in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.025 M $\text{KIO}_3(aq)$ by diluting a 0.500 M stock solution.

(h) What volume of 0.500 M stock solution, in mL, would be required to prepare 100. mL of 0.025 M $\text{KIO}_3(aq)$?

(i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student spilled some of the stock solution while transferring it to the volumetric flask and continues to fill it to the line with distilled water, would the resulting solution have a concentration that is greater than, less than, or equal to the intended concentration? Justify your answer.

The $\text{IO}_3^-(aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (I) are represented below.



(j) Identify the mass spectrum that represents the isotopes of iodine. Justify your choice.

Iodine can react with a certain element, M. The successive ionization energy values for element M are given below.

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	Ionization Energy (kJ/mol)
First	578
Second	1817
Third	2745
Fourth	11,578
Fifth	14,842

(k) What is the charge of the ion formed by element M in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element M would form with iodine?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct hybridization.

· sp^3

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response lists **all** the intermolecular forces present in both compounds.

· London dispersion forces, dipole-dipole forces

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response provides both a correct answer and a valid justification equivalent to the following.

· Disagree. The student has the claim reversed. Compound Y does not experience hydrogen bonding, while compound W does, resulting in stronger total intermolecular forces for compound W and therefore a higher boiling point.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. Only a second order reaction will give a linear plot of $1/[X]$ versus time with a positive slope.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the rate law that is consistent with the graph.

· Rate = $k[X]^2$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both of the following criteria.

☐

The response calculates the correct value of k consistent with the rate law in (e).

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$$k = \text{slope} = \frac{(20-10) M^{-1}}{(15-5) \text{ day}} = 1 M^{-1} \text{ day}^{-1}$$

☐

The response includes the correct units.

$$M^{-1} \text{ day}^{-1}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The rate constant would be larger at higher temperature. Accordingly, the slope of the line from the second experiment will be steeper than that from the first experiment.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a calculation of the correct volume.

$$M_1 V_1 = M_2 V_2$$

$$0.500 M \times V_1 = 0.025 M \times 100.0 \text{ mL}$$

$$V_1 = 5.0 \text{ mL}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Less than. Compared to the intended quantities, there would be fewer moles of solute (KIO_3) dissolved in the same volume of solution.

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Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the identification of spectrum A and a justification equivalent to one of the following.

· Spectrum A. The weighted average molar mass of I is $126.91 \frac{\text{g}}{\text{mol}}$, indicating that there is a higher abundance of I – 127 than I – 126.

OR

· $(127 \text{ amu} \times 0.9) + (126 \text{ amu} \times 0.1) = 126.9 \text{ amu}$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct charge and a justification equivalent to the following.

· $3+$. The large increase between the third and fourth ionization energy indicates that the element has three valence electrons.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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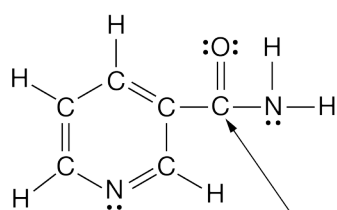
The response gives a correct formula consistent with part (k).

· MI_3

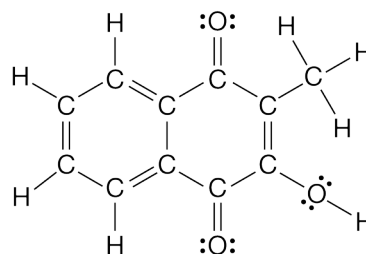
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7. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound W and Compound Z. Structures of the compounds are shown below.



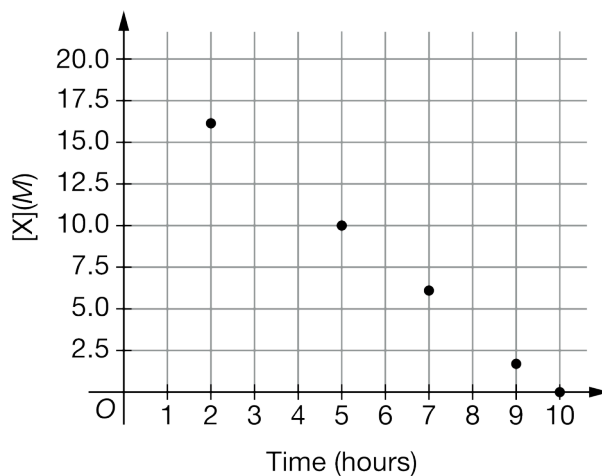
Compound W
(molar mass = 122 g/mol)



Compound Z
(molar mass = 186 g/mol)

- (a) Using VSEPR theory, predict the O — C — N bond angle indicated by the arrow in the diagram of compound W.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound Z has a higher boiling point than compound W. The student claims that compound Z has a greater boiling point because it has greater London dispersion forces than compound W. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

A different compound found in food, compound X is known to decompose over time. To study the reaction, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting [X] versus time.



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(d) The student proposes that the reaction is zero order with respect to compound X. Do you agree or disagree? Justify your answer using the data in the graph above.

(e) Write the rate law expression for the decomposition of compound X that is consistent with the graph above.

(f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.

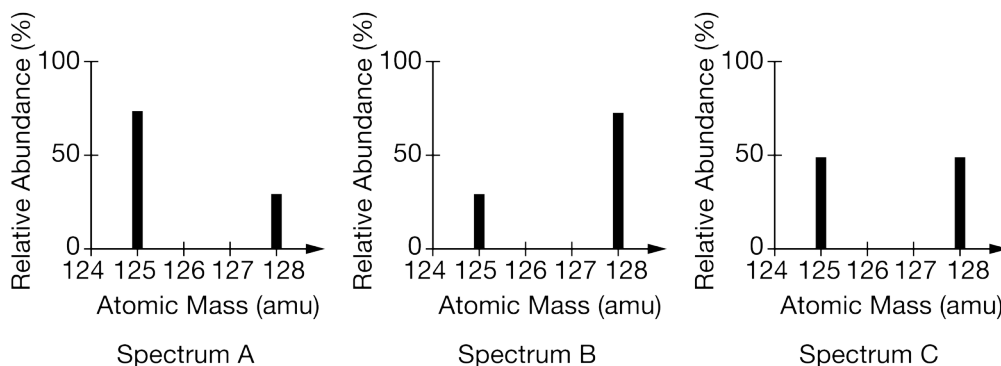
(g) The student claims that if the experiment were conducted again at a higher temperature, the value for the rate constant, k , would increase. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the higher temperature conditions.

A student titrates a solution of compound X to determine the concentration of X in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.010 M $\text{KIO}_3(aq)$ by diluting a 0.040 M stock solution.

(h) What volume of 0.040 M stock solution, in mL, would be required to prepare 100. mL of 0.010 M $\text{KIO}_3(aq)$?

(i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student filled the volumetric flask with distilled water but the bottom of the meniscus was significantly above the 100.0 mL mark, would the resulting solution concentration be greater than, less than, or equal to what was intended? Justify your answer.

The $\text{IO}_3^-(aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (I) are represented below.



(j) Identify the mass spectrum that most likely represents the isotopes of iodine. Justify your choice.

Iodine can also react with a certain element, M. The successive ionization energy values for element M are given below.

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	Ionization Energy (kJ/mol)
First	419
Second	3051
Third	4419
Fourth	5878
Fifth	7975

(k) What is the charge of the ion formed by element M in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element M would form with iodine?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct angle.

· 120°

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response lists **all** of the intermolecular forces present in both compounds.

· London dispersion forces, dipole-dipole forces, hydrogen bonding

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response provides both a correct answer and a valid justification equivalent to the following.

· Agree. The greater number of atoms in compound Z leads to a larger, more polarizable electron cloud than that of compound W. Thus, Z has stronger London dispersion forces and a higher boiling point than W has. (Both compounds have hydrogen bonding.)

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. Only a zeroth order reaction will give a linear plot of $[X]$ versus time with a negative slope.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the rate law that is consistent with the graph.

· $\text{Rate} = k[X]^0 = k$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both of the following criteria.

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- ☐ The response calculates the correct value of k consistent with the rate law in (e).
· $k = -\text{slope} = -\frac{(0-10)M}{(10-5)\text{hr}} = 2\text{ M hr}^{-1}$
- ☐ The response includes the correct units.
· M hr^{-1}

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. The rate constant would be larger at higher temperature. Accordingly, the slope of the line from the second experiment will be steeper than that from the first experiment.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a calculation of the correct volume.

$$\cdot M_1 V_1 = M_2 V_2$$

$$0.040\text{ M} \times V_1 = 0.010\text{ M} \times 100.0\text{ mL}$$

$$V_1 = 25\text{ mL}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- ☐ The response provides both the correct answer and a valid justification equivalent to the following.
· Less than. Compared to the intended quantities, there would be the same number of moles of solute (

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KIO_3) dissolved in a greater volume of solution.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the identification of spectrum B and a justification equivalent to one of the following.

· Spectrum B. The weighted average molar mass of I is $126.91 \frac{\text{g}}{\text{mol}}$, indicating that there is a higher abundance of I – 128 than I – 125.

OR

· $(128 \text{ amu} \times 0.7) + (125 \text{ amu} \times 0.3) = 127 \text{ amu}$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct charge and a justification equivalent to the following.

· $1+$. The large increase between the first and second ionization energy indicates that the element has one valence electron.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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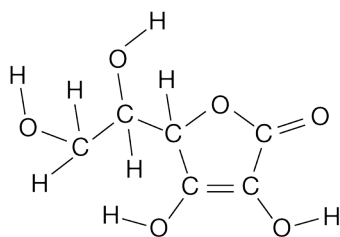
The response gives the correct formula consistent with part (k).

· MnI

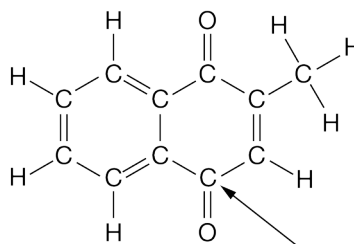
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8. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates properties of two compounds found in food, labeled Compound X and Compound Y. Structures of the compounds are shown below.



Compound X
(molar mass = 176 g/mol)

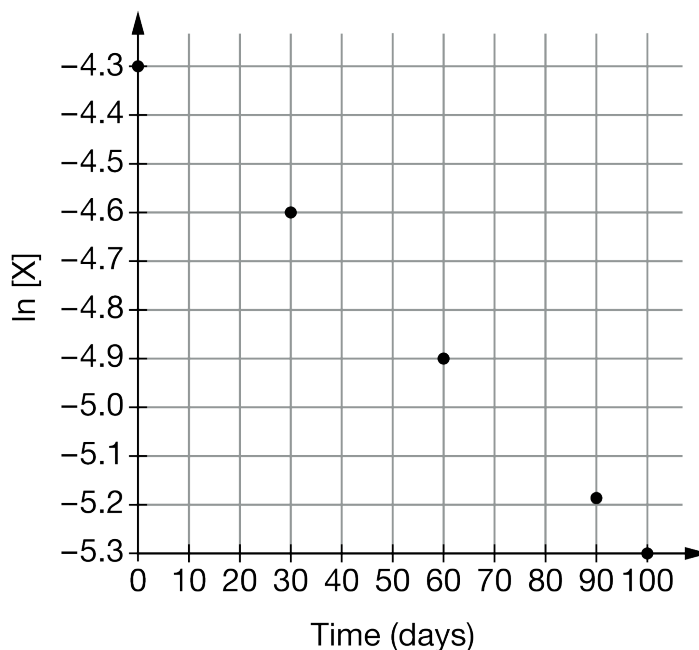


Compound Y
(molar mass = 172 g/mol)

- (a) Identify the hybridization of the valence orbitals of the carbon atom indicated with the arrow in the diagram of compound Y.
- (b) List all of the type(s) of intermolecular force(s) that the two compounds have in common.
- (c) Compound X has a greater boiling point than compound Y. The student claims that the difference is because compound X exhibits hydrogen bonding, but compound Y does not. Do you agree or disagree with this claim? Justify your answer in terms of the types and relative strengths of intermolecular forces in each substance.

To show that compound X present in food is lost over time, a student monitors the molar concentration of compound X in a solution at constant temperature. After collecting and analyzing the data, the following linear graph was created by plotting $\ln [X]$ versus time.

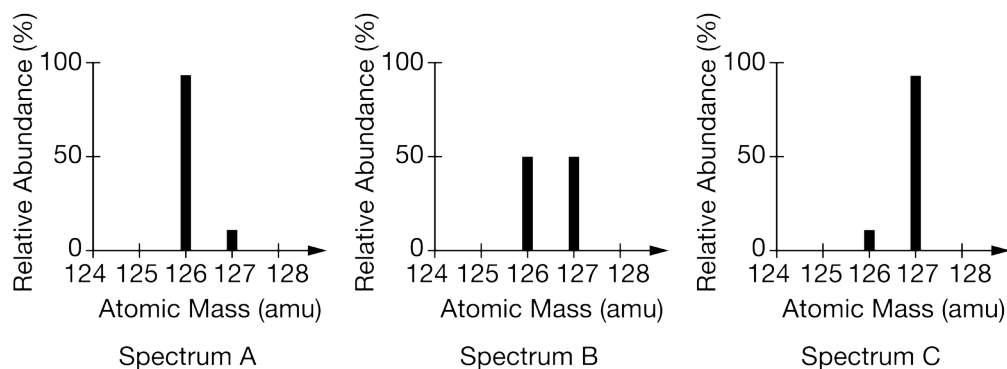
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- (d) The student proposes that the reaction is first order with respect to compound X. Do you agree or disagree? Justify your answer using the data in the graph above.
- (e) Write the rate law expression for the decomposition of compound X that is consistent with the graph above.
- (f) Determine the value of the rate constant, k , for the reaction. Include units with your answer.
- (g) The student claims that if the experiment were conducted again at a lower temperature, the value for the rate constant, k , would decrease. Do you agree or disagree with the student's claim? Justify your answer by comparing the slope of the original line above to the slope that would result from the lower temperature conditions.
- A student titrates a solution of compound X to determine the concentration of X in the solution. In order to conduct the titration, the student needs to prepare 100. mL of 0.050 M $\text{KIO}_3(aq)$ by diluting a 0.500 M stock solution.
- (h) What volume of 0.500 M stock solution, in mL, would be required to prepare 100. mL of 0.050 M $\text{KIO}_3(aq)$?
- (i) To prepare the diluted solution, the student uses a graduated cylinder to measure the volume of stock solution calculated in part (h) and adds it to a 100.0 mL volumetric flask, followed by the addition of distilled water. If the student fills the volumetric flask with distilled water, but the bottom of the meniscus is significantly below the 100.0 mL mark, would the resulting solution concentration be greater than, less than, or equal to the intended concentration? Justify your answer.

The $\text{IO}_3^-(aq)$ ion can be reduced to form elemental iodine. Three possible mass spectra for a sample of atomic iodine (I) are represented below.

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(j) Identify the mass spectrum that represents the isotopes of iodine. Justify your choice.

Iodine can react with a certain element, M . The successive ionization energy values for element M are given below.

	Ionization Energy (kJ/mol)
First	738
Second	1451
Third	7733
Fourth	10,542
Fifth	13,636

(k) What is the charge of the ion formed by element M in its compounds? Justify your answer using the data in the table.

(l) Given the charge determined in part (k), what is the formula for the compound that element M would form with iodine?

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct hybridization.

· sp^2

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response lists **all** the intermolecular forces present in both compounds.

· London dispersion forces, dipole-dipole forces

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both a correct answer and a valid justification equivalent to the following.

· Agree. The hydrogen bonding in compound X makes the total of all intermolecular forces stronger compared to compound Y, resulting in a higher boiling point.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. Only a first order reaction will give a linear plot of $\ln[X]$ versus time with a negative slope.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response provides the rate law that is consistent with the graph.

$$\text{Rate} = k[\text{X}]^1 = k[\text{X}]$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both of the following criteria.

- ☐ The response calculates the correct value of k consistent with the rate law in (e).
 $k = -\text{slope} = -\frac{-4.6 - (-4.3)}{(30 - 0) \text{ day}} = 0.010 \text{ day}^{-1}$
- ☐ The response includes the correct units.
 day^{-1}

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. The rate constant would be smaller at lower temperature. Accordingly, the slope of the line from the second experiment will be less steep than that from the first experiment.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a calculation of the correct volume.

$$M_1 V_1 = M_2 V_2$$

$$0.500 M \times V_1 = 0.050 M \times 100.0 \text{ mL}$$

$$V_1 = 10. \text{ mL}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Greater than. Compared to the intended quantities, there would be the same number of moles of solute (KIO_3) dissolved in a smaller volume of solution.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the identification of spectrum C and a justification equivalent to one of the following.

· Spectrum C. The weighted average molar mass of I is $126.91 \frac{\text{g}}{\text{mol}}$, indicating that there is a higher abundance of I – 127 than I – 126.

OR

$$(127 \text{ amu} \times 0.9) + (126 \text{ amu} \times 0.1) = 126.9 \text{ amu}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct charge and a justification equivalent to the following.

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· 2+. The large increase between the second and third ionization energy indicates that the element has two valence electrons.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

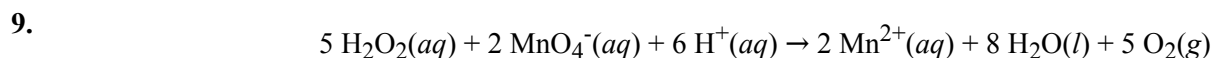


0

1

The response gives a correct formula consistent with part (k).

· MI_2

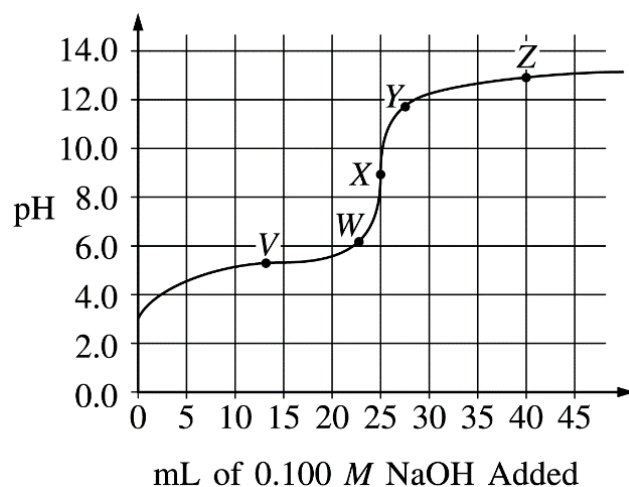


In a titration experiment, $\text{H}_2\text{O}_2(aq)$ reacts with aqueous $\text{MnO}_4^-(aq)$ as represented by the equation above. The dark purple KMnO_4 solution is added from a buret to a colorless, acidified solution of $\text{H}_2\text{O}_2(aq)$ in an Erlenmeyer flask. (Note: At the end point of the titration, the solution is a pale pink color.)

Which of the following best describes what happens to the pH of the H_2O_2 solution as the titration proceeds?

- (A) The +2 charge on the manganese ions maintains the acidity of the solution.
- (B) The production of water dilutes the solution, making it basic.
- (C) As H^+ ions are consumed, the solution becomes less acidic and the pH increases. ✓
- (D) As H^+ ions are consumed, the solution becomes less acidic and the pH decreases.

The graph below shows the titration curve that results when 100. mL of 0.0250 M acetic acid is titrated with 0.100 M NaOH.



Chapter 11 Part 4: Titration

10. Which of the following indicators is the best choice for this titration?

(A)

<u>Indicator</u>	<u>pH Range of Color Change</u>
Methyl orange	3.2 - 4.4

(B)

<u>Indicator</u>	<u>pH Range of Color Change</u>
Methyl red	3.2 - 4.4

(C)

<u>Indicator</u>	<u>pH Range of Color Change</u>
Bromothymol	6.1 - 7.6

(D)

<u>Indicator</u>	<u>pH Range of Color Change</u>
Phenolphthalein	8.2 - 10.0



(E)

<u>Indicator</u>	<u>pH Range of Color Change</u>
Alizarin	11.0 - 12.4

11. What part of the curve corresponds to the optimum buffer action for the acetic acid/acetate ion pair?

(A) Point *V* 

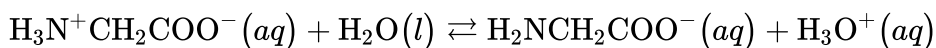
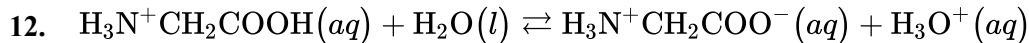
(B) Point *X*

(C) Point *Z*

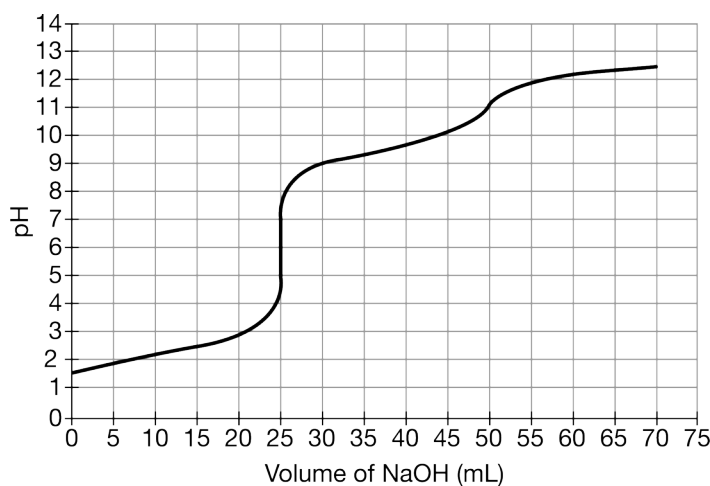
(D) Along all of section *WY*

(E) Along all of section *YZ*

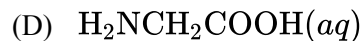
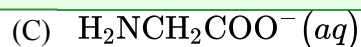
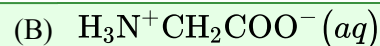
Chapter 11 Part 4: Titration



The stepwise dissociation of the amino acid glycine is represented by the chemical equations above. A student titrates a sample of glycine dissolved in dilute acid with $0.100\text{ M NaOH}(aq)$. The data are plotted on the following graph.

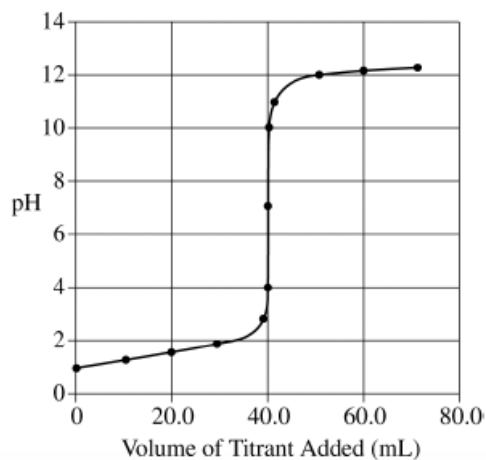


Based on the data, which of the following species has the highest concentration in an aqueous solution of glycine with a pH of 7?



Chapter 11 Part 4: Titration

13.

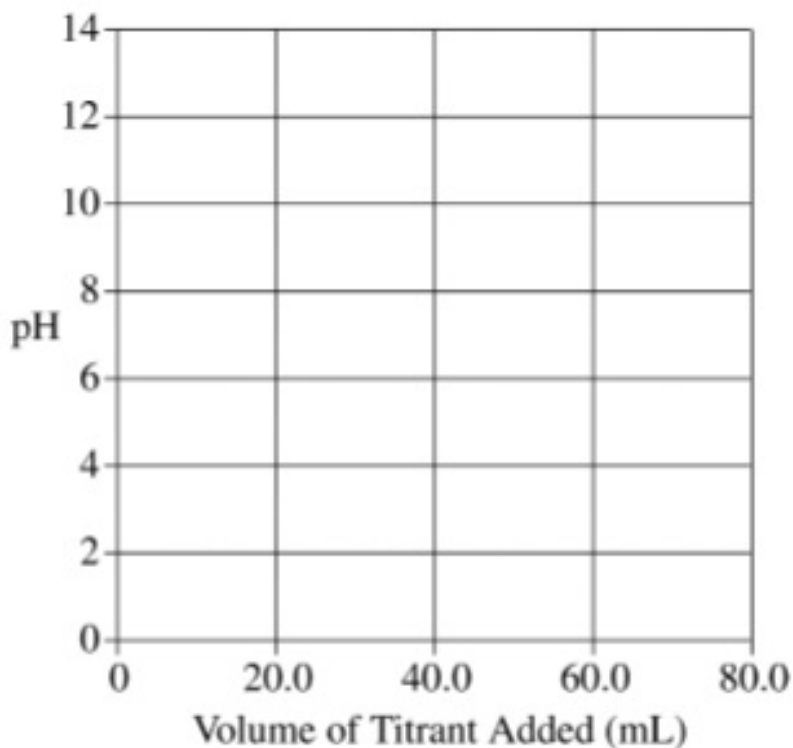


A solution of 0.100 M HCl and a solution of 0.100 M NaOH are prepared. A 40.0 mL sample of one of the solutions is added to a beaker and then titrated with the other solution. A pH electrode is used to obtain the data that are plotted in the titration curve shown above.

- Identify the solution that was initially added to the beaker. Explain your reasoning.
- On the titration curve above, circle the point that corresponds to the equivalence point.
- At the equivalence point, how many moles of titrant have been added?
- The same titration is to be performed again, this time using an indicator. Use the information in the table below to select the best indicator for the titration. Explain your choice.

Indicator	pH Range of Color Change
Methyl violet	0 – 1.6
Methyl red	4 – 6
Alizarin yellow	10 – 12

- What is the difference between the equivalence point of a titration and the end point of a titration?
- On the grid provided on the next page, sketch the titration curve that would result if the solutions in the beaker and buret were reversed (i.e., if 40.0 mL of the solution used in the buret in the previous titration were titrated with the solution that was in the beaker).

Chapter 11 Part 4: Titration**Part A**

1 point is earned for the correct identification with rationale.

The solution in the beaker was the 0.100 *M* HCl because the initial pH was 1 (the pH of 0.100 *M* HCl).



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct identification with rationale.

The solution in the beaker was the 0.100 *M* HCl because the initial pH was 1 (the pH of 0.100 *M* HCl).

Part B

1 point is earned for the correct choice of point.

The point with coordinates (40.0, 7) is circled.

Chapter 11 Part 4: Titration

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct choice of point.

The point with coordinates (40.0, 7) is circled.

Part C

1 point is earned for the correct numerical answer.

$$0.0400 \text{ L} \times \frac{0.100 \text{ mol NaOH}}{1.00 \text{ L}} = 0.00400 \text{ mol NaOH}$$

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct numerical answer.

$$0.0400 \text{ L} \times \frac{0.100 \text{ mol NaOH}}{1.00 \text{ L}} = 0.00400 \text{ mol NaOH}$$

Part D

1 point is earned for the correct selection of indicator.

1 point is earned for the explanation.

Methyl red would be best because its color change will occur closest to the equivalence point (when the pH changes from about 4 to 10).

✓

0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct selection of indicator.

1 point is earned for the explanation.

Methyl red would be best because its color change will occur closest to the equivalence point (when the pH changes from

Chapter 11 Part 4: Titration

about 4 to 10).

Part E

1 point is earned for each correct definition (for a maximum of 2 points).

The equivalence point in a titration occurs when the number of moles of titrant added is exactly sufficient to react completely with the number of moles of the titrated species present in the sample being titrated.

The end point of a titration is the point in a titration at which the indicator undergoes its color change.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for each correct definition (for a maximum of 2 points).

The equivalence point in a titration occurs when the number of moles of titrant added is exactly sufficient to react completely with the number of moles of the titrated species present in the sample being titrated.

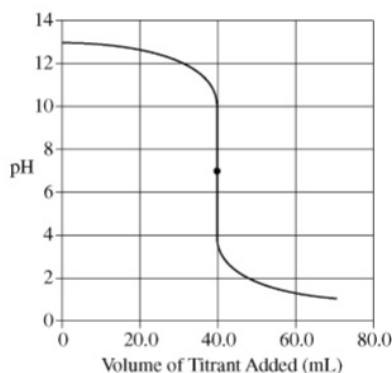
The end point of a titration is the point in a titration at which the indicator undergoes its color change.

Part F

1 point is earned for starting between pH 12 and 14 and for finishing below pH 2.

1 point is earned for locating the equivalence point at pH 7 and volume 40.0 mL.

1 point is earned for the overall shape of the curve.



0	1	2	3
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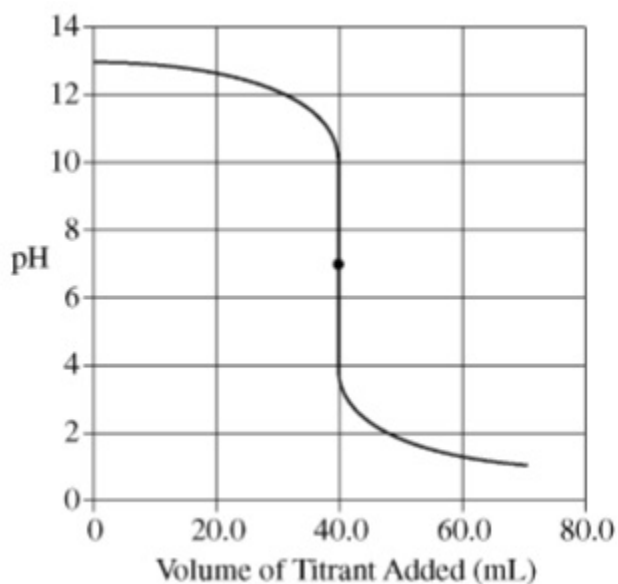
The student response earns three of the following points:

Chapter 11 Part 4: Titration

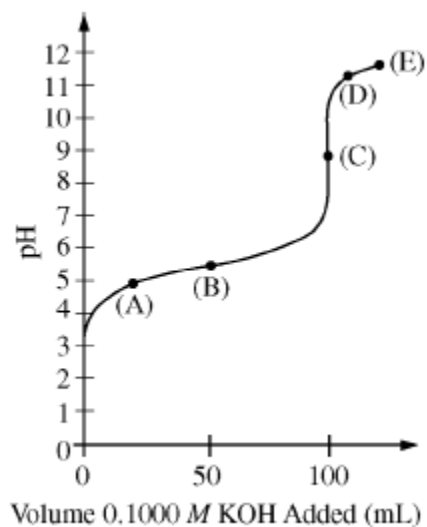
1 point is earned for starting between pH 12 and 14 and for finishing below pH 2.

1 point is earned for locating the equivalence point at pH 7 and volume 40.0 mL.

1 point is earned for the overall shape of the curve.



14. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.



The point at which the moles of the added strong base are equal to the moles of the weak acid initially present

Chapter 11 Part 4: Titration

(A) A

(B) B

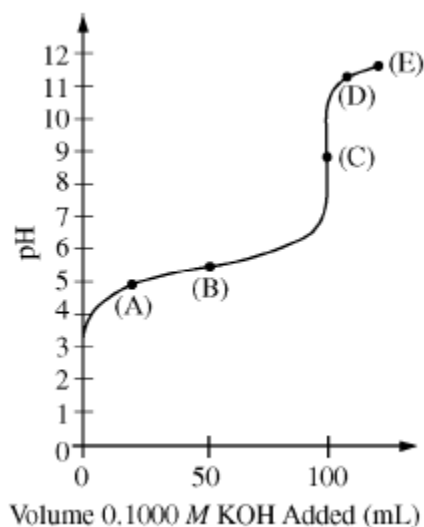
(C) C



(D) D

(E) E

15. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.



The point at which the pH is closest to that of the strong base being added

(A) A

(B) B

(C) C

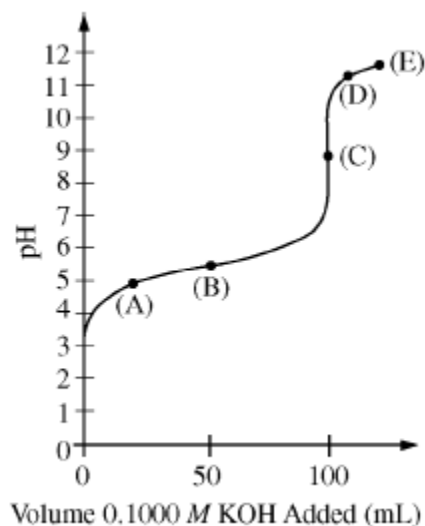
(D) D

(E) E



Chapter 11 Part 4: Titration

16. A solution of a weak monoprotic acid is titrated with a solution of a strong base, KOH. Consider the points labeled (A) through (E) on the titration curve that results, as shown below.

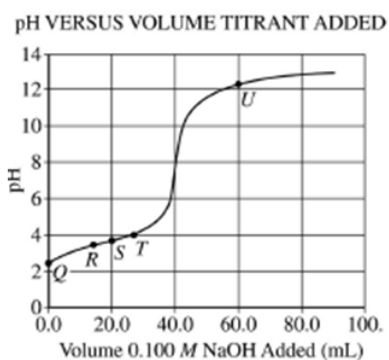


The point at which the concentrations of the weak acid and its conjugate base are approximately equal

- (A) A
(B) B
(C) C
(D) D
(E) E



Refer to the following.



A 50.0 mL sample of an acid, HA, of unknown molarity is titrated, and the pH of the resulting solution is measured with a meter and graphed as a function of the volume of 0.100 M NaOH added.

Chapter 11 Part 4: Titration

17. A student carries out the same titration but uses an indicator instead of a pH meter. If the indicator changes color slightly past the equivalence point, what will the student obtain for the calculated concentration of the acid?

(A) Slightly less than 0.0800 *M*

(B) Slightly more than 0.0800 *M* ✓

(C) Slightly less than 0.125 *M*

(D) Slightly more than 0.125 *M*

18. At point *R* in the titration, which of the following species has the highest concentration?

(A) HA ✓

(B) A^-

(C) H_3O^+

(D) OH^-

19. At which point on the titration curve is $[A^-]$ closest to twice that of $[HA]$?

(A) *R*

(B) *S*

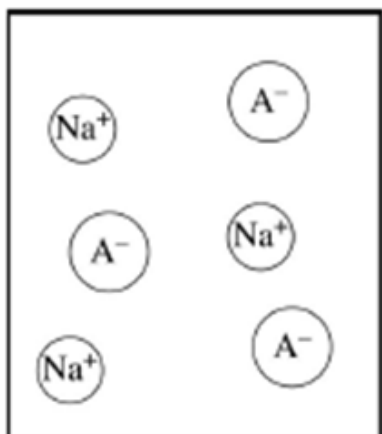
(C) *T* ✓

(D) *U*

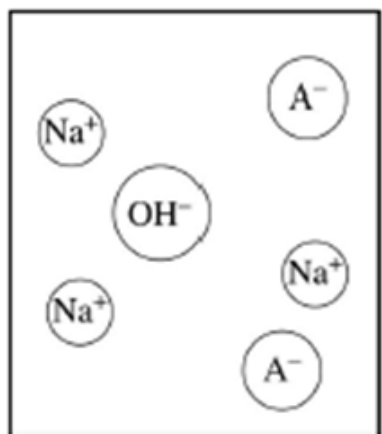
20. Which of the following is the best particulate representation of the species (other than H_2O) that are present in significant concentrations in the solution at point *U* in the titration?

Chapter 11 Part 4: Titration

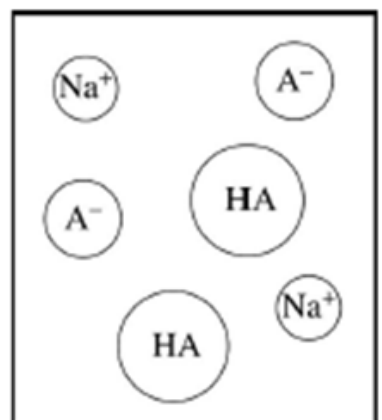
(A)



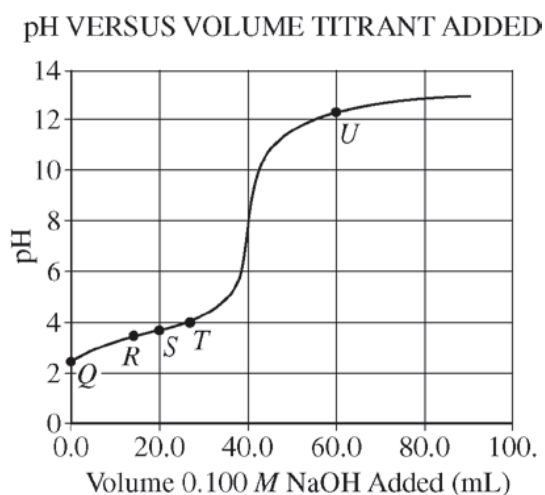
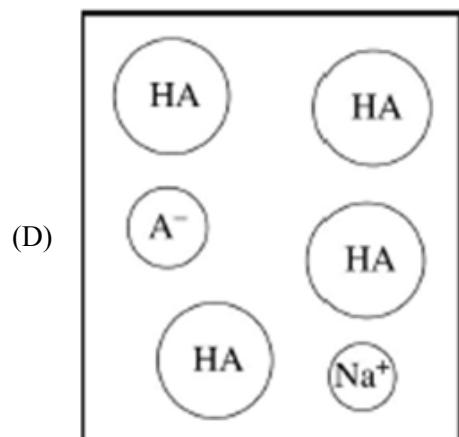
(B)



(C)



Chapter 11 Part 4: Titration



A 50.0 mL sample of an acid, HA, of unknown molarity is titrated, and the pH of the resulting solution is measured with a pH meter and graphed as a function of the volume of 0.100 M NaOH added.

21. A student carries out the same titration, but uses an indicator instead of a pH meter. If the indicator changes color slightly past the equivalence point, what will the student obtain for the calculated concentration of the acid?
- (A) Slightly less than 0.0800 M
- (B) Slightly more than 0.0800 M ✓
- (C) Slightly less than 0.125 M
- (D) Slightly more than 0.125 M
22. At point R in the titration, which of the following species has the highest concentration?

Chapter 11 Part 4: Titration

(A) HA



(B) A^-

(C) H_3O^+

(D) OH^-

23. At which point on the titration curve is $[A^-]$ closest to twice that of $[HA]$?

(A) *R*

(B) *S*

(C) *T*

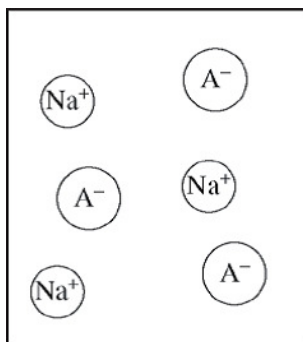


(D) *U*

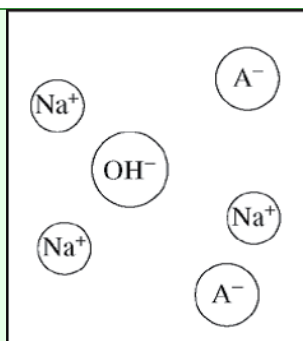
24. Which of the following is the best particulate representation of the species (other than H_2O) that are present in significant concentrations in the solution at point *U* in the titration?

Chapter 11 Part 4: Titration

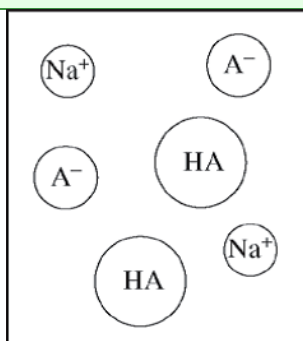
(A)



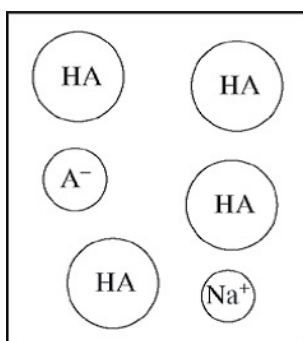
(B)



(C)



(D)



Chapter 11 Part 4: Titration

25. A student pipetted five 25.00-milliliter samples of hydrochloric acid and transferred each sample to an Erlenmeyer flask, diluted it with distilled water, and added a few drops of phenolphthalein to each. Each sample was then titrated with a sodium hydroxide solution to the appearance of the first permanent faint pink color. The following results were obtained.

Volumes of NaOH Solution

First Sample.....35.22 mL

Second Sample.....36.14 mL

Third Sample.....36.13 mL

Fourth Sample36.15 mL

Fifth Sample.....36.12 mL

Which of the following is the most probable explanation for the variation in the student's results?

- (A) The burette was not rinsed with NaOH solution.
- (B) The student misread a 5 for a 6 on the burette when the first sample was titrated.
- (C) A different amount of water was added to the first sample.
- (D) The pipette was not rinsed with the HCl solution. ✓
- (E) The student added too little indicator to the first sample.

26.
$$\text{C}_9\text{H}_8\text{O}_4(aq) + \text{NaOH}(aq) \rightleftharpoons \text{C}_9\text{H}_7\text{O}_4^-(aq) + \text{Na}^+(aq) + \text{H}_2\text{O}(l)$$

The reaction represented above occurs when 2.00×10^{-4} mol of pure acetylsalicylic acid, $\text{C}_9\text{H}_8\text{O}_4$, is completely dissolved in 15.00 mL of water in a flask and titrated to the equivalence point with 0.100 M NaOH(aq). Which of the following statements about the titration is true at the equivalence point?

- (A) $[\text{C}_9\text{H}_8\text{O}_4]$ is greater than $[\text{C}_9\text{H}_7\text{O}_4^-]$.
- (B) $[\text{C}_9\text{H}_8\text{O}_4]$ is less than $[\text{C}_9\text{H}_7\text{O}_4^-]$. ✓
- (C) $[\text{C}_9\text{H}_8\text{O}_4]$ is equal to $[\text{C}_9\text{H}_7\text{O}_4^-]$.
- (D) The relationship between $[\text{C}_9\text{H}_8\text{O}_4]$ and $[\text{C}_9\text{H}_7\text{O}_4^-]$ cannot be determined without additional information.

Chapter 11 Part 4: Titration

27.

Acid Solution	Volume of NaOH Added (mL)
A	40
B	75
C	115
D	200

To maximize the yield in a certain manufacturing process, a solution of a weak monoprotic acid that has a concentration between 0.20 M and 0.30 M is required. Four 100. mL samples of the acid at different concentrations are each titrated with a 0.20 M NaOH solution. The volume of NaOH needed to reach the end point for each sample is given in the table above. Which solution is the most suitable to maximize the yield?

(A) Solution A

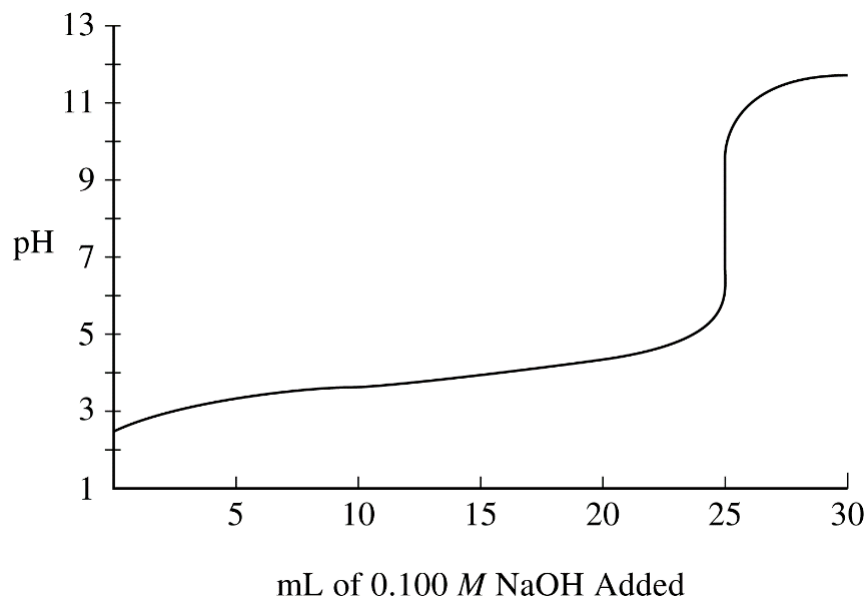
(B) Solution B

(C) Solution C

(D) Solution D

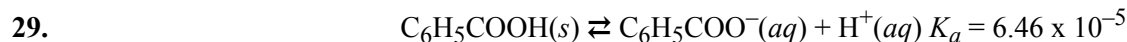


28.



An unknown acid is dissolved in 25 mL of water and titrated with 0.100 M NaOH. The results are shown in the titration curve above. Which of the following could be the unknown acid?

Chapter 11 Part 4: Titration

(A) Fluoroacetic acid, $pK_a = 2.6$ (B) Glycolic acid, $pK_a = 3.8$ ✓(C) Propanoic acid, $pK_a = 4.9$ (D) Hypochlorous acid, $pK_a = 7.5$ (E) Boric acid, $pK_a = 9.3$ 

Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, dissociates in water as shown in the equation above. A 25.0 mL sample of an aqueous solution of pure benzoic acid is titrated using standardized 0.150 M NaOH.

- a. After addition of 15.0 mL of the 0.150 M NaOH, the pH of the resulting solution is 4.37. Calculate each of the following.
 - i. $[\text{H}^+]$ in the solution
 - ii. $[\text{OH}^-]$ in the solution
 - iii. The number of moles of NaOH added
 - iv. The number of moles of $\text{C}_6\text{H}_5\text{COO}^-(aq)$ in the solution
 - v. The number of moles of $\text{C}_6\text{H}_5\text{COOH}$ in the solution
- b. State whether the solution at the equivalence point of the titration is acidic, basic, or neutral. Explain your reasoning.

In a different titration, a 0.7529 g sample of a mixture of solid $\text{C}_6\text{H}_5\text{COOH}$ and solid NaCl is dissolved in water and titrated with 0.150 M NaOH. The equivalence point is reached when 24.78 mL of the base solution is added.

- c. Calculate each of the following.
 - i. The mass, in grams, of benzoic acid in the solid sample
 - ii. The mass percentage of benzoic acid in the solid sample

Part A

1 point is earned for the correct answer.

$$[\text{H}^+] = 10^{-4.37} \text{ M} = 4.3 \times 10^{-5} \text{ M}$$

1 point is earned for the correct answer.

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14} \text{ M}^2}{4.3 \times 10^{-5} \text{ M}} = 2.3 \times 10^{-10} \text{ M}$$

Chapter 11 Part 4: Titration

1 point is earned for the correct answer.

$$\text{mol OH}^- = 0.0150 \text{ L} \times 0.150 \text{ mol L}^{-1} = 2.25 \times 10^{-3} \text{ mol}$$

1 point is earned for the correct answer.

mol OH[−] added = mol C₆H₅COO[−](aq) generated, thus

$$\text{mol C}_6\text{H}_5\text{COO}^-(\text{aq}) \text{ in solution} = 2.25 \times 10^{-3} \text{ mol}$$

1 point is earned for the correct molarity.

$$K_a = \frac{[\text{H}^+][\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]} \Rightarrow [\text{C}_6\text{H}_5\text{COOH}] = \frac{[\text{H}^+][\text{C}_6\text{H}_5\text{COO}^-]}{K_a}$$

$$[\text{C}_6\text{H}_5\text{COOH}] = \frac{(4.3 \times 10^{-5} \text{ M}) \times \frac{2.25 \times 10^{-3} \text{ mol}}{0.040 \text{ L}}}{6.46 \times 10^{-5}} = 3.7 \times 10^{-2} \text{ M}$$

1 point is earned for the correct answer.

$$\text{thus, mol C}_6\text{H}_5\text{COOH} = (0.040 \text{ L})(3.7 \times 10^{-2} \text{ M}) = 1.5 \times 10^{-3} \text{ mol}$$

OR

1 point is earned for the correct molarity.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]}$$

$$\Rightarrow \text{pH} - \text{p}K_a = \log [\text{C}_6\text{H}_5\text{COO}^-] - \log [\text{C}_6\text{H}_5\text{COOH}]$$

$$\Rightarrow \log [\text{C}_6\text{H}_5\text{COOH}] = \log [\text{C}_6\text{H}_5\text{COO}^-] - (\text{pH} - \text{p}K_a)$$

$$= \log \left(\frac{2.25 \times 10^{-3} \text{ mol}}{0.040 \text{ L}} \right) - (4.37 - 4.190)$$

$$= -1.25 - 0.18 = -1.43$$

$$\Rightarrow [\text{C}_6\text{H}_5\text{COOH}] = 10^{-1.43} = 3.7 \times 10^{-2} \text{ M}$$

1 point is earned for the correct answer.

$$\text{thus, mol C}_6\text{H}_5\text{COOH} = (0.040 \text{ L})(3.7 \times 10^{-2} \text{ M}) = 1.5 \times 10^{-3} \text{ mol}$$



0	1	2	3	4	5	6
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The student response earns six of the following points:

1 point is earned for the correct answer.

$$[\text{H}^+] = 10^{-4.37} \text{ M} = 4.3 \times 10^{-5} \text{ M}$$

Chapter 11 Part 4: Titration

1 point is earned for the correct answer.

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14} M^2}{4.3 \times 10^{-5} M} = 2.3 \times 10^{-10} M$$

1 point is earned for the correct answer.

$$\text{mol OH}^- = 0.0150 \text{ L} \times 0.150 \text{ mol L}^{-1} = 2.25 \times 10^{-3} \text{ mol}$$

1 point is earned for the correct answer.

mol OH[−] added = mol C₆H₅COO[−](aq) generated, thus

$$\text{mol C}_6\text{H}_5\text{COO}^-(aq) \text{ in solution} = 2.25 \times 10^{-3} \text{ mol}$$

1 point is earned for the correct molarity.

$$K_a = \frac{[\text{H}^+][\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]} \Rightarrow [\text{C}_6\text{H}_5\text{COOH}] = \frac{[\text{H}^+][\text{C}_6\text{H}_5\text{COO}^-]}{K_a}$$

$$[\text{C}_6\text{H}_5\text{COOH}] = \frac{(4.3 \times 10^{-5} M) \times \frac{2.25 \times 10^{-3} \text{ mol}}{0.040 \text{ L}}}{6.46 \times 10^{-5}} = 3.7 \times 10^{-2} M$$

1 point is earned for the correct answer.

$$\text{thus, mol C}_6\text{H}_5\text{COOH} = (0.040 \text{ L})(3.7 \times 10^{-2} M) = 1.5 \times 10^{-3} \text{ mol}$$

OR

1 point is earned for the correct molarity.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{C}_6\text{H}_5\text{COO}^-]}{[\text{C}_6\text{H}_5\text{COOH}]}$$

$$\Rightarrow \text{pH} - \text{p}K_a = \log [\text{C}_6\text{H}_5\text{COO}^-] - \log [\text{C}_6\text{H}_5\text{COOH}]$$

$$\Rightarrow \log [\text{C}_6\text{H}_5\text{COOH}] = \log [\text{C}_6\text{H}_5\text{COO}^-] - (\text{pH} - \text{p}K_a)$$

$$= \log\left(\frac{2.25 \times 10^{-3} \text{ mol}}{0.040 \text{ L}}\right) - (4.37 - 4.190)$$

$$= -1.25 - 0.18 = -1.43$$

$$\Rightarrow [\text{C}_6\text{H}_5\text{COOH}] = 10^{-1.43} = 3.7 \times 10^{-2} M$$

1 point is earned for the correct answer.

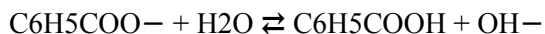
$$\text{thus, mol C}_6\text{H}_5\text{COOH} = (0.040 \text{ L})(3.7 \times 10^{-2} M) = 1.5 \times 10^{-3} \text{ mol}$$

Part B

1 point is earned for the prediction and the explanation.

At the equivalence point the solution is basic due to the presence of C₆H₅COO[−] (the conjugate base of the weak acid) that hydrolyzes to produce a basic solution as represented below.

Chapter 11 Part 4: Titration

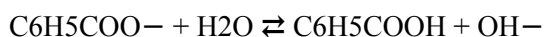


0	1
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The student response earns one of the following points:

1 point is earned for the prediction and the explanation.

At the equivalence point the solution is basic due to the presence of $\text{C}_6\text{H}_5\text{COO}^-$ (the conjugate base of the weak acid) that hydrolyzes to produce a basic solution as represented below.



Part C

1 point is earned for the correct answer.

$$\text{mol C}_6\text{H}_5\text{COOH} = (0.02478 \text{ L}) \times (0.150 \text{ mol OH}^- \text{ L}^{-1}) \times \frac{1 \text{ mol C}_6\text{H}_5\text{COOH}}{1 \text{ mol OH}^-}$$

$$= 3.72 \times 10^{-3} \text{ mol C}_6\text{H}_5\text{COOH}$$

$$\text{mass C}_6\text{H}_5\text{COOH} = 3.72 \times 10^{-3} \text{ mol C}_6\text{H}_5\text{COOH} \times \frac{122 \text{ g C}_6\text{H}_5\text{COOH}}{1 \text{ mol C}_6\text{H}_5\text{COOH}}$$

$$= 0.453 \text{ g C}_6\text{H}_5\text{COOH}$$

1 point is earned for the correct answer.

$$\text{mass \% C}_6\text{H}_5\text{COOH} = \frac{0.453 \text{ g C}_6\text{H}_5\text{COOH}}{0.7529 \text{ g}} \times 100$$

$$= 60.2\%$$



0	1	2
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The student response earns two of the following points:

1 point is earned for the correct answer.

$$\text{mol C}_6\text{H}_5\text{COOH} = (0.02478 \text{ L}) \times (0.150 \text{ mol OH}^- \text{ L}^{-1}) \times \frac{1 \text{ mol C}_6\text{H}_5\text{COOH}}{1 \text{ mol OH}^-}$$

$$= 3.72 \times 10^{-3} \text{ mol C}_6\text{H}_5\text{COOH}$$

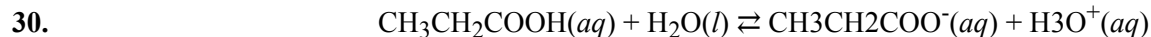
$$\text{mass C}_6\text{H}_5\text{COOH} = 3.72 \times 10^{-3} \text{ mol C}_6\text{H}_5\text{COOH} \times \frac{122 \text{ g C}_6\text{H}_5\text{COOH}}{1 \text{ mol C}_6\text{H}_5\text{COOH}}$$

$$= 0.453 \text{ g C}_6\text{H}_5\text{COOH}$$

Chapter 11 Part 4: Titration

1 point is earned for the correct answer.

$$\text{mass } \% \text{ C}_6\text{H}_5\text{COOH} = \frac{0.453 \text{ g C}_6\text{H}_5\text{COOH}}{0.7529 \text{ g}} \times 100$$
$$= 60.2\%$$



Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, is a carboxylic acid that reacts with water according to the equation above. At 25°C the pH of a 50.0 mL sample of 0.20 M $\text{CH}_3\text{CH}_2\text{COOH}$ is 2.79.

- Identify a Brønsted-Lowry conjugate acid-base pair in the reaction. Clearly label which is the acid and which is the base.
- Determine the value of K_a for propanoic acid at 25°C .
- For each of the following statements, determine whether the statement is true or false. In each case, explain the reasoning that supports your answer.
 - The pH of a solution prepared by mixing the 50.0 mL sample of 0.20 M $\text{CH}_3\text{CH}_2\text{COOH}$ with a sample 50.0 mL 0.20 M NaOH is 7.00.
 - If the pH of a hydrochloric acid solution is the same as the pH of a propanoic acid solution, then the molar concentration of the hydrochloric acid solution must be less than the molar concentration of the propanoic acid solution.

A student is given the task of determining the concentration of a propanoic acid solution of unknown concentration. A 0.173 M NaOH solution is available to use as the titrant. The student uses a 25.00 mL volumetric pipet to deliver the propanoic acid solution to a clean, dry flask. After adding an appropriate indicator to the flask, the student titrates the solution with the 0.173 M NaOH, reaching the end point after 20.52 mL of the base solution has been added.

- Calculate the molarity of the propanoic acid solution.
- The student is asked to redesign the experiment to determine the concentration of a butanoic acid solution instead of a propanoic acid solution. For butanoic acid the value of $\text{p}K_a$ is 4.83. The student claims that a different indicator will be required to determine the equivalence point of the titration accurately. Based on your response to part (b), do you agree with the student's claim? Justify your answer.

Part A

1 point is earned for writing (or naming) either of the Brønsted-Lowry conjugate acid-base pairs with a clear indication of which is the acid and which is the base.

$\text{CH}_3\text{CH}_2\text{COOH}$ and $\text{CH}_3\text{CH}_2\text{COO}^-$

acid base

OR

Chapter 11 Part 4: Titration

H₃O⁺ and H₂O

acid base



0	1
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The student response earns one of the following points:

1 point is earned for writing (or naming) either of the Brønsted-Lowry conjugate acid-base pairs with a clear indication of which is the acid and which is the base.

CH₃CH₂COOH and CH₃CH₂COO⁻

acid base

OR

H₃O⁺ and H₂O

acid base

Part B

1 point is earned for correctly solving for [H₃O⁺].

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} = 10^{-2.79} = 1.6 \times 10^{-3} \text{ M}$$

1 point is earned for the *K_a* expression for propanoic acid

OR 1 point is earned for substituting values into the *K_a* expression.

$$[\text{CH}_3\text{CH}_2\text{COO}^-] = [\text{H}_3\text{O}^+]$$

AND

$$[\text{CH}_3\text{CH}_2\text{COOH}] = 0.20 \text{ M} - [\text{H}_3\text{O}^+], \text{ OR } [\text{CH}_3\text{CH}_2\text{COOH}] \approx 0.20 \text{ M (state or assume that } [\text{H}_3\text{O}^+] \ll 0.20 \text{ M)}$$

1 point is earned for correctly solving for the value of *K_a*.

$$K_a = \frac{[\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{CH}_2\text{COOH}]} = \frac{(1.6 \times 10^{-3} \text{ M})^2}{0.20 \text{ M}} = 1.3 \times 10^{-5}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

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1 point is earned for correctly solving for $[H_3O^+]$.

$$[H_3O^+] = 10^{-pH} = 10^{-2.79} = 1.6 \times 10^{-3} M$$

1 point is earned for the K_a expression for propanoic acid

OR 1 point is earned for substituting values into the K_a expression.

$$[CH_3CH_2COO^-] = [H_3O^+]$$

AND

$$[CH_3CH_2COOH] = 0.20 M - [H_3O^+], \text{ OR } [CH_3CH_2COOH] \approx 0.20 M \text{ (state or assume that } [H_3O^+] \ll 0.20 M \text{)}$$

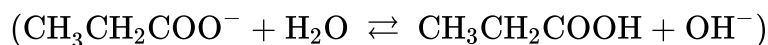
1 point is earned for correctly solving for the value of K_a .

$$K_a = \frac{[CH_3CH_2COO^-][H_3O^+]}{[CH_3CH_2COOH]} = \frac{(1.6 \times 10^{-3} M)^2}{0.20 M} = 1.3 \times 10^{-5}$$

Part C

- 1 point is earned for noting that the statement is false AND providing a supporting explanation.

False. The conjugate base of a weak acid undergoes hydrolysis (see equation below) at equivalence to form a solution with a $pH > 7$.



- 1 point is earned for noting that the statement is true and providing a supporting explanation.

True. HCl is a strong acid that ionizes completely. Fewer moles of HCl are needed to produce the same $[H_3O^+]$ as the propanoic acid solution, which only partially ionizes.

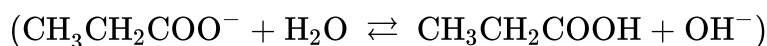


0	1	2
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The student response earns two of the following points:

- 1 point is earned for noting that the statement is false AND providing a supporting explanation.

False. The conjugate base of a weak acid undergoes hydrolysis (see equation below) at equivalence to form a solution with a $pH > 7$.



- 1 point is earned for noting that the statement is true and providing a supporting explanation.

True. HCl is a strong acid that ionizes completely. Fewer moles of HCl are needed to produce the same $[H_3O^+]$ as the propanoic acid solution, which only partially ionizes.

Part D

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1 point is earned for correctly calculating the number of moles of acid that reacted at the equivalence point.

1 point is earned for the correct molarity of acid.

Let x = moles of propanoic acid

$$\text{then } x = (0.02052 \text{ L NaOH}) \times \frac{0.173 \text{ mol NaOH}}{1 \text{ L NaOH}} \times \frac{1 \text{ mol acid}}{1 \text{ mol NaOH}}$$

$$= 3.55 \times 10^{-3} \text{ mol propanoic acid}$$

$$\frac{3.55 \times 10^{-3} \text{ mol acid}}{0.02500 \text{ L acid}} = 0.142 \text{ M}$$

OR

Since $\text{CH}_3\text{CH}_2\text{COOH}$ is monoprotic and, at the equivalence point, moles H^+ = moles OH^- , then

$$MAVA = MBVB$$

$$M_A = \frac{M_B V_B}{V_A} = \frac{(0.173 \text{ M NaOH})(20.52 \text{ mL NaOH})}{25.00 \text{ mL acid}} = 0.142 \text{ M}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly calculating the number of moles of acid that reacted at the equivalence point.

1 point is earned for the correct molarity of acid.

Let x = moles of propanoic acid

$$\text{then } x = (0.02052 \text{ L NaOH}) \times \frac{0.173 \text{ mol NaOH}}{1 \text{ L NaOH}} \times \frac{1 \text{ mol acid}}{1 \text{ mol NaOH}}$$

$$= 3.55 \times 10^{-3} \text{ mol propanoic acid}$$

$$\frac{3.55 \times 10^{-3} \text{ mol acid}}{0.02500 \text{ L acid}} = 0.142 \text{ M}$$

OR

Since $\text{CH}_3\text{CH}_2\text{COOH}$ is monoprotic and, at the equivalence point, moles H^+ = moles OH^- , then

$$MAVA = MBVB$$

$$M_A = \frac{M_B V_B}{V_A} = \frac{(0.173 \text{ M NaOH})(20.52 \text{ mL NaOH})}{25.00 \text{ mL acid}} = 0.142 \text{ M}$$

Part E

1 point is earned for disagreeing with the student's claim and making a valid justification using pK_a , K_a , pH arguments.

Chapter 11 Part 4: Titration

1 point is earned for numerically comparing either: the two pK_a values, the two K_a values, or the two pH values at the equivalence point.

Disagree with the student's claim

From part (b) above, pK_a for propanoic acid is $\log(1.3 \times 10^{-5}) = 4.89$. Because 4.83 is so close to 4.89, the pH at the equivalence point in the titration of butanoic acid should be close enough to the pH in the titration of propanoic acid to make the original indicator appropriate for the titration of butanoic acid.



0	1	2
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The student response earns two of the following points:

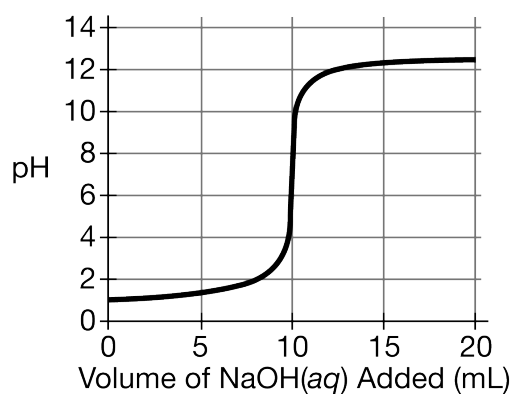
1 point is earned for disagreeing with the student's claim and making a valid justification using pK_a , K_a , pH arguments.

1 point is earned for numerically comparing either: the two pK_a values, the two K_a values, or the two pH values at the equivalence point.

Disagree with the student's claim

From part (b) above, pK_a for propanoic acid is $\log(1.3 \times 10^{-5}) = 4.89$. Because 4.83 is so close to 4.89, the pH at the equivalence point in the titration of butanoic acid should be close enough to the pH in the titration of propanoic acid to make the original indicator appropriate for the titration of butanoic acid.

31.



A 20.0 mL sample of a 0.125 M monoprotic acid solution is titrated with a $\text{NaOH}(aq)$ solution of unknown concentration. Based on the titration curve above, what is the molar concentration of the $\text{NaOH}(aq)$ solution?

(A) 0.063 M

(B) 0.125 M

(C) 0.250 M

(D) 0.500 M



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32.

Acid	Concentration of Acid (<i>M</i>)	Volume of Sample (mL)	Volume of Standardized NaOH(<i>aq</i>) Added to Reach Equivalence Point (mL)
HClO ₄ (<i>aq</i>)	0.100	25.0	24.0
HClO(<i>aq</i>)	0.200	25.0	?

Samples of two different acids are titrated using the same standardized solution of NaOH(*aq*). According to the data in the table above, how many milliliters of the standardized NaOH(*aq*) are required to reach the equivalence point when the 25.0 mL sample of 0.200 *M* HClO(*aq*) is titrated?

(A) 12.0 mL

(B) 24.0 mL

(C) 48.0 mL

(D) It cannot be determined without knowing the values of K_a for HClO₄(*aq*) and HClO(*aq*).

33. Potassium hydrogen phthalate, KHP, is used as a primary standard for determining the concentration of a solution of NaOH by titration. If the KHP has not been dried before weighing, the calculated molarity of the NaOH would be

(A) higher than the actual value, since water is included in the apparent mass of KHP



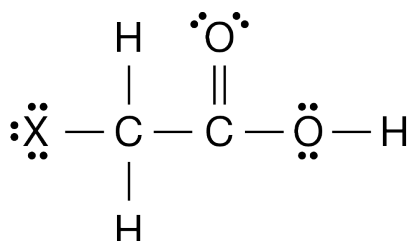
(B) higher than the actual value, since the presence of water requires a larger volume of titrant

(C) lower than the actual value, since NaOH absorbs water

(D) unaffected, since KHP is a strong acid

(E) unaffected, since water is routinely added before the titration

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The structure of haloacetic acids, XCH_2COOH (where X is either F , Cl , Br , or I), is shown above. The dissociation constants and molar masses of four haloacetic acids are listed in the table below.

Acid	$\text{p}K_a$	K_a	Molar Mass(g/mol)
Fluoroacetic acid	2.59	2.57×10^{-3}	78.0
Chloroacetic acid	2.87	1.35×10^{-3}	94.5
Bromoacetic acid	2.90	1.26×10^{-3}	138.9
Iodoacetic acid	3.18	6.61×10^{-4}	185.9

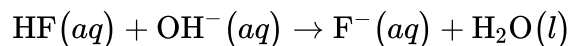
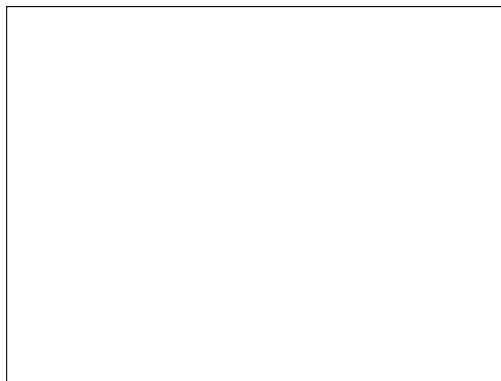
34. A student titrates 10.0 mL samples of 1.0 M solutions of each of the haloacetic acids with a standard solution of NaOH . Which of the following statements correctly predicts the volume of $\text{NaOH}(aq)$ needed to reach the equivalence point?
- (A) Fluoroacetic acid will need the smallest volume of $\text{NaOH}(aq)$ to reach the equivalence point.
- (B) Iodoacetic acid will need the smallest volume of $\text{NaOH}(aq)$ to reach the equivalence point.
- (C) All of the acids will need the same volume of $\text{NaOH}(aq)$ to reach the equivalence point. ✓
- (D) All of the haloacetic acids are weak; therefore none will reach an equivalence point.

Chapter 11 Part 4: Titration

35. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to HF and the titration of $\text{HF}(aq)$.

- (a) In the following box, draw the complete Lewis electron-dot diagram of the HF molecule.

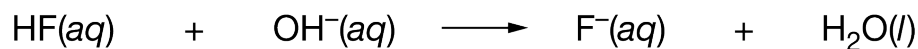
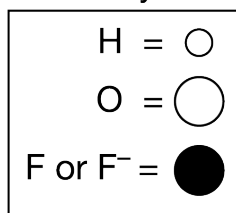


A student performed a titration of $\text{HF}(aq)$ with $\text{NaOH}(aq)$. The net ionic equation for the neutralization reaction that occurs during the titration is shown above.

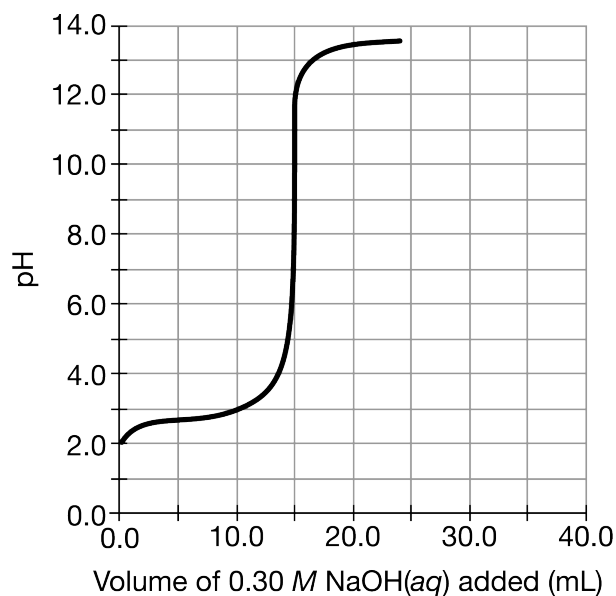
- (b) Identify both of the Brønsted-Lowry conjugate acid-base pairs in the neutralization reaction above. For each pair, label the acid and the base.
- (c) In terms of the species in the reaction, explain what occurs during the reaction that identifies it as an acid-base reaction.
- (d) The equation can be written in particulate form. Using the following key, draw the correct products above their formulas.

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Key



A student titrated 30. mL of 0.15 M $\text{HF}(aq)$ with 0.30 M $\text{NaOH}(aq)$. The results are plotted on the following graph.



- (e) How many moles of HF are in 30. mL of 0.15 M $\text{HF}(aq)$?
- (f) Draw an X on the horizontal axis of the graph to show the amount of $\text{NaOH}(aq)$ that had been added when the equivalence point was reached.
- (g) The graph above shows the results of the titration of the 30. mL of 0.15 M $\text{HF}(aq)$ with 0.30 M $\text{NaOH}(aq)$. On the same graph, draw the curve for the titration of 60. mL of 0.15 M $\text{HF}(aq)$ with 0.30 M $\text{NaOH}(aq)$.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. Electron pairs can be

Chapter 11 Part 4: Titration

shown as a line or as two dots.



0	1
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The drawing meets both of the criteria below:

- ☐ The response shows an **H** atom and an **F** atom joined by a single bond.
- ☐ The response shows three nonbonding pairs of electrons on the **F** atom and none on the **H** atom.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets all of the criteria below:

- ☐ **HF** and **F⁻** is identified as one Bronsted-Lowry acid-base pairs.
- ☐ **HF** is identified as an acid and **F⁻** is identified as a base.
- ☐ **OH⁻** and **H₂O** is identified as one Bronsted-Lowry acid-base pairs.
- ☐ **OH⁻** is identified as a base and **H₂O** is identified as an acid.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates one of the criteria below:

- ☐ **HF** donates a proton and becomes **F⁻**
- ☐ **HF** donates a proton to **OH⁻**)
- ☐ **OH⁻** accepts a proton and becomes **H₂O** (OR that **OH⁻** accepts a proton from **HF**) .

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2
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The drawings include both of the following:

- ☐ A dark (or shaded) circle is drawn above the F^- .
- ☐ An H_2O molecule (drawn as a large unshaded circle with two small unshaded circles attached to it) is drawn above the H_2O .

(e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives the following calculation (or a variation):

$$\frac{0.15 \text{ mol HF}}{1000 \text{ mL}} \times 30. \text{ mL} = 0.0045 \text{ mol (or } 4.5 \times 10^{-3} \text{ mol) (unit not required)}$$

(f)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows an X on the horizontal axis at 15.0 mL OR on the curve at 15.0 mL . (If the X is drawn on the curve, it does not need to be at the correct pH .)

Part (g)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



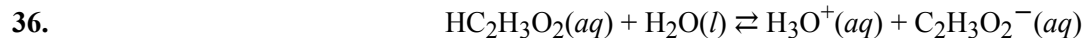
0	1	2
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The drawing meets all of the criteria below:

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- ☐ The curve is shaped like a titration curve.
- ☐ The curve starts at a **pH** essentially the same as the first curve.
- ☐ The roughly vertical part of the curve is at a volume of 30.0 mL .

Chapter 11 Part 4: Titration



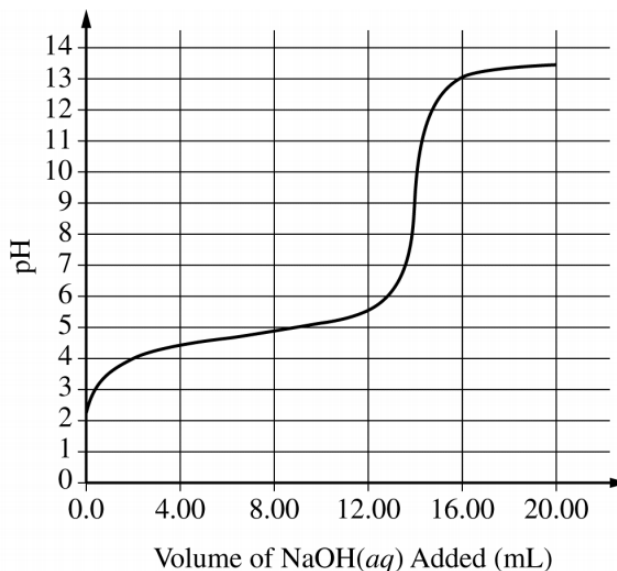
The dissociation of ethanoic acid, $\text{HC}_2\text{H}_3\text{O}_2(aq)$, is represented above. A student is given the task of determining the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using two different experimental procedures.

- a. The student is first asked to prepare 100.0 mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using a 2.000 M standard solution.
 - i. Calculate the volume, in mL, of 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ the student needs to prepare 100.0 mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$.
 - ii. Describe the procedure the student should use to prepare 100. mL of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using appropriate equipment selected from the list below. Assume that the student uses appropriate safety equipment.
 - 100 mL beaker
 - 100 mL graduated cylinder
 - 100 mL volumetric flask
 - Eye dropper
 - 500 mL wash bottle filled with distilled water
 - 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ in a 50 mL buret
- b. Using a pH probe, the student determines that the pH of 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ is 2.92.
 - i. Using the pH value, calculate the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$
 - ii. Calculate the percent dissociation of ethanoic acid in 0.115 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$

In a separate experimental procedure, the student titrates 10.0 mL of the 2.000 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ with an $\text{NaOH}(aq)$ solution of unknown concentration. The student monitors the pH during the titration. The following titration curve was created using the experimental data presented in the table.

Chapter 11 Part 4: Titration

Volume of NaOH(aq) Added (mL)	pH
0.00	2.23
2.00	3.99
4.00	4.37
6.00	4.65
8.00	4.90
10.00	5.17
12.00	5.55
14.00	9.35
16.00	13.04
18.00	13.31
20.00	13.46



- Write the balanced net ionic equation for the reaction that occurs when $\text{HC}_2\text{H}_3\text{O}_2(aq)$ and $\text{NaOH}(aq)$ are combined.
- Calculate the molar concentration of the $\text{NaOH}(aq)$ solution.
- Explain how the student can estimate the value of K_a for $\text{HC}_2\text{H}_3\text{O}_2(aq)$ using the titration curve.

Part A

3 point(s) maximum

Part (i)

1 point is earned for the correct volume.

$$M_i V_i = M_f V_f$$

$$V_i = \frac{(0.115M)(100.0 \text{ mL})}{2.000M} = 5.75$$

mL

Part (ii)

1 point is earned for dispensing from the buret.

Use the buret to deliver 5.75 mL of 2.000 M $\text{HC}_2\text{H}_3\text{O}_2$ to the 100 mL volumetric flask.

1 point is earned for diluting the solution to the calibration mark of the volumetric flask.

Then add distilled water from the wash bottle to the flask (adding the last few drops with an eyedropper) until the volume of liquid in the flask is at the calibration mark

Chapter 11 Part 4: Titration



0	1	2	3
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Student response earns 3 of the following 3 point(s)

3 point(s) maximum

Part (i)

1 point is earned for the correct volume.

$$M_i V_i = M_f V_f$$

$$V_i = \frac{(0.115M)(100.0\text{ mL})}{2.000M} = 5.75$$

mL

Part (ii)

1 point is earned for dispensing from the buret.

Use the buret to deliver 5.75 mL of 2.000M HC₂H₃O₂ to the 100 mL volumetric flask.

1 point is earned for diluting the solution to the calibration mark of the volumetric flask.

Then add distilled water from the wash bottle to the flask (adding the last few drops with an eyedropper) until the volume of liquid in the flask is at the calibration mark

Part B

3 point(s) maximum

Part (i)

1 point is earned for correct conversion of pH to [H₃O⁺].

$$\text{pH} = 2.92 \Rightarrow [\text{H}_3\text{O}^+] = 10^{-2.92} = 0.0012\text{ M}$$

1 point is earned for a value of K_a consistent with the student's value of [H₃O⁺].

$$K_a = \frac{[\text{H}^3\text{O}^+][\text{C}_2\text{H}_3\text{O}_2^-]}{[\text{HC}_2\text{H}_3\text{O}_2]}$$

Since [H₃O⁺] = [C₂H₃O₂⁻], then

$$K_a = \frac{(0.0012)(0.0012)}{(0.115 - 0.0012)} = \frac{(0.0012)^2}{(0.114)} = 1.3 \times 10^{-5}$$

Chapter 11 Part 4: Titration

Part (ii)

1 point is earned for the correct percent dissociation.

Percent dissociation =

$$\frac{[\text{C}_2\text{H}_3\text{O}_2^-]}{[\text{HC}_2\text{H}_3\text{O}_2]_0} = \frac{0.0012}{0.115} \times 100 = 1.0\%$$



0	1	2	3
---	---	---	---

Student response earns 3 of the following 3 point(s)

3 point(s) maximum

Part (i)

1 point is earned for correct conversion of pH to [H₃O⁺].

$$\text{pH} = 2.92 \Rightarrow [\text{H}_3\text{O}^+] = 10^{-2.92} = 0.0012 \text{ M}$$

1 point is earned for a value of K_a consistent with the student's value of [H₃O⁺].

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{C}_2\text{H}_3\text{O}_2^-]}{[\text{HC}_2\text{H}_3\text{O}_2]}$$

Since [H₃O⁺] = [C₂H₃O₂⁻], then

$$K_a = \frac{(0.0012)(0.0012)}{(0.115 - 0.0012)} = \frac{(0.0012)^2}{(0.114)} = 1.3 \times 10^{-5}$$

Part (ii)

1 point is earned for the correct percent dissociation.

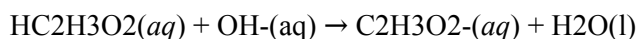
Percent dissociation =

$$\frac{[\text{C}_2\text{H}_3\text{O}_2^-]}{[\text{HC}_2\text{H}_3\text{O}_2]_0} = \frac{0.0012}{0.115} \times 100 = 1.0\%$$

Part C

1 point(s) maximum

1 point is earned for the correct equation.



Chapter 11 Part 4: Titration

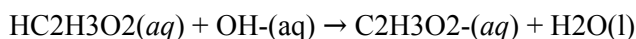


0	1
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Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for the correct equation.



Part D

2 point(s) maximum

1 point is earned for determining the moles of acid.

From the pH curve, the equivalence point occurs at 14.0 mL.

$$10 \text{ mL} \times \frac{2.000 \text{ mol HC}_2\text{H}_3\text{O}_2}{1000 \text{ mL}} = 0.0200 \text{ mol HC}_2\text{H}_3\text{O}_2(aq)$$

1 point is earned for determining the molar concentration of the base.

$$0.0200 \text{ mol HC}_2\text{H}_3\text{O}_2(aq) \times \frac{1 \text{ mol NaOH}}{1 \text{ mol HC}_2\text{H}_3\text{O}_2} = 0.0200 \text{ mol NaOH}$$

$$\frac{0.0200 \text{ mol NaOH}}{0.0140 \text{ L solution}} = 1.43 \text{ M NaOH}(aq)$$



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for determining the moles of acid.

From the pH curve, the equivalence point occurs at 14.0 mL.

$$10 \text{ mL} \times \frac{2.000 \text{ mol HC}_2\text{H}_3\text{O}_2}{1000 \text{ mL}} = 0.0200 \text{ mol HC}_2\text{H}_3\text{O}_2(aq)$$

1 point is earned for determining the molar concentration of the base.

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$$0.0200 \text{ mol HC}_2\text{H}_3\text{O}_2(aq) \times \frac{1 \text{ mol NaOH}}{1 \text{ mol HC}_2\text{H}_3\text{O}_2} = 0.0200 \text{ mol NaOH}$$

$$\frac{0.0200 \text{ mol NaOH}}{0.0140 \text{ L solution}} = 1.43 \text{ M NaOH}(aq)$$

Part E

1 point(s) maximum

1 point is earned for a correct explanation (numerical explanation not required).

At the half-equivalence point (~7.0 mL) the pH of the solution is equal to the pKa of the acid. The antilog of the negative pH is equal to the value of Ka.



0	1
---	---

Student response earns 1 of the following 1 point(s)

1 point(s) maximum

1 point is earned for a correct explanation (numerical explanation not required).

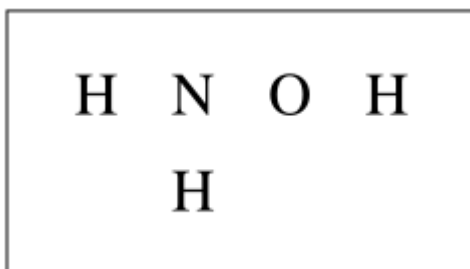
At the half-equivalence point (~7.0 mL) the pH of the solution is equal to the pKa of the acid. The antilog of the negative pH is equal to the value of Ka.

Chapter 11 Part 4: Titration

37. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about hydroxylamine, NH_2OH .

- (a) Complete the Lewis electron-dot diagram of the NH_2OH molecule in the box below, including all bonding and nonbonding valence electrons.

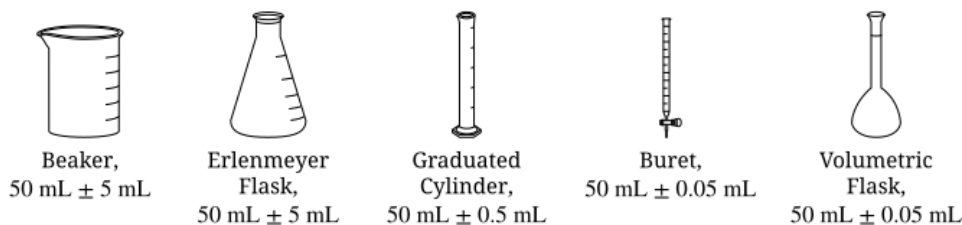


- (b) Hydroxylamine, $\text{NH}_2\text{OH}(aq)$, is a weak base. Write the balanced net ionic equation for the reaction that occurs when $\text{NH}_2\text{OH}(aq)$ and $\text{HCl}(aq)$ are combined.

A student is preparing to titrate a $\text{NH}_2\text{OH}(aq)$ solution with $\text{HCl}(aq)$. To prepare for the titration, the student places the NH_2OH solution into an Erlenmeyer flask.

- (c) The titration procedure instructs the student to prepare 100.00mL of a 0.1000M HCl solution. The stock solution available is 0.5604M HCl . Calculate the volume, in liters, of stock solution required for the dilution.

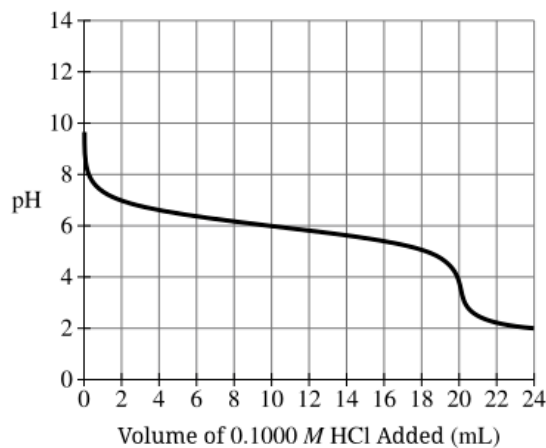
- (d) Using the following choices, identify the glassware that should be used to deliver the $\text{HCl}(aq)$ with the greatest precision. Circle your choice and provide a brief justification.



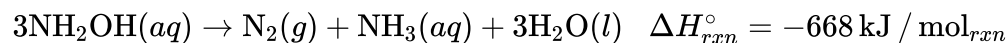
- (e) The student titrates a 25.00mL sample of a NH_2OH solution with 0.1000M HCl . The student monitors the pH during the experiment and creates the following titration curve. Selected data are given in the table.

Chapter 11 Part 4: Titration

Volume of HCl Added (mL)	pH
0.00	9.67
2.00	7.00
6.00	6.40
8.00	6.21
10.00	6.03
12.00	5.85
14.00	5.66
18.00	5.08
20.00	3.44
24.00	2.01



- (i) Based on the data from the titration experiment, estimate the value of pK_b for NH_2OH .
- (ii) Calculate the molar concentration of the 25.00 mL sample of $\text{NH}_2\text{OH}(aq)$.
- (iii) In a second trial, using a different 25.00 mL sample of NH_2OH solution, the volume of HCl used to reach the equivalence point is greater than in the first trial. The student incorrectly claims that the 0.5604 M stock solution of HCl was used in the second trial instead of the diluted solution. Explain why the student's claim is incorrect.



At elevated temperatures, $\text{NH}_2\text{OH}(aq)$ undergoes decomposition according to the equation shown.

- (f) If 8.00g of NH_2OH decomposes according to the equation above, how much heat, in kJ, is produced?

Part A

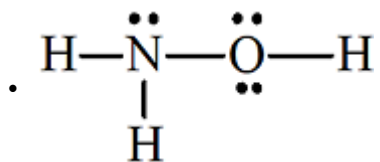
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

Chapter 11 Part 4: Titration



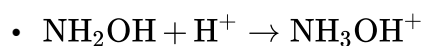
Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

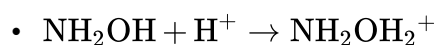


0	1
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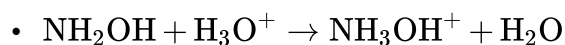
The response provides a correct balanced equation.



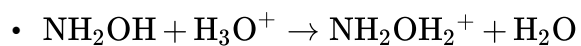
OR



OR



OR



Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

- $V_1 = \frac{M_2 V_2}{M_1} = \frac{(0.1000 \text{ M})(0.10000 \text{ L})}{0.5604 \text{ M}} = 0.01784 \text{ L}$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 4: Titration



0	1
---	---

The response provides the correct answer and a valid justification.

- The buret is the best choice to dispense volume precisely, because it can be used to dispense a variety of different volumes to the nearest 0.05 mL.

Part E(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct calculated value.

- The halfway point in the titration occurs when approximately 10 mL of titrant is added. At the half-equivalence point, the pH of the solution equals the pK_a of the conjugate acid, NH_3OH^+ . According to the table, the pH is 6.03, so $pK_a + pK_b = 14.00$

$$pK_b = 14.00 - 6.03 = 7.97$$

OR

- Based on the graph, the pH at the half-equivalence point is approximately 6.0, so $pK_a + pK_b = 14.00$.

$$pK_b = 14.0 - 6.0 = 8.0$$

Part E(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **both** of the following criteria.

☐

The response provides the correct volume.

- 20.00 mL of 0.1000 M HCl is required to reach the equivalence point.

Chapter 11 Part 4: Titration

- ☐ The response provides the correct calculated value of concentration.

$$\begin{aligned} & 0.02000 \text{ L HCl} \times \frac{0.1000 \text{ mol HCl}}{1 \text{ L}} \times \frac{1 \text{ mol NH}_2\text{OH}}{1 \text{ mol HCl}} = 0.002000 \text{ mol NH}_2\text{OH} \\ & \text{Concentration of NH}_2\text{OH} = \frac{0.002000 \text{ mol}}{0.02500 \text{ L}} = 0.08000 \text{ M} \end{aligned}$$

Part E(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

- If a more concentrated HCl solution is used to titrate the same amount of base, the volume of acid required to reach the equivalence point would be smaller, not greater. Therefore, using the stock solution could not have caused this error in the second trial.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets **both** of the following criteria.

- ☐ The response provides the correct calculated value of moles of reaction.

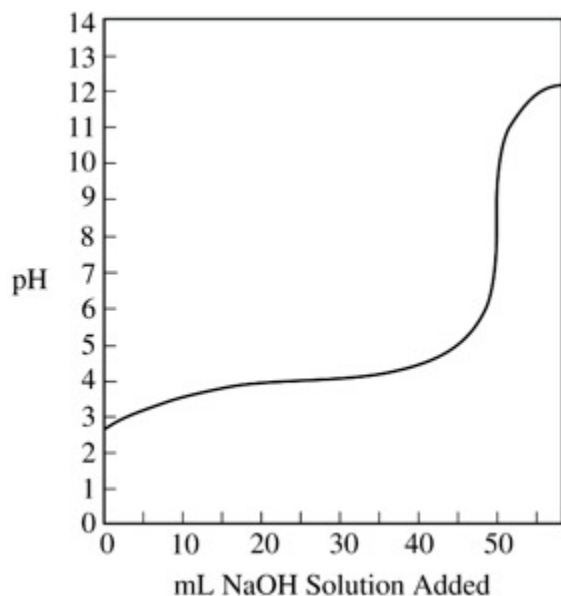
$$8.00 \text{ g NH}_2\text{OH} \times \frac{1 \text{ mol NH}_2\text{OH}}{33.03 \text{ g NH}_2\text{OH}} \times \frac{1 \text{ mol}_{rxn}}{3 \text{ mol NH}_2\text{OH}} = 0.0807 \text{ mol}_{rxn}$$

- ☐ The response provides the correct calculated value of heat produced.

$$0.0807 \text{ mol}_{rxn} \times \frac{-668 \text{ kJ}}{1 \text{ mol}_{rxn}} = -53.9 \text{ kJ. Therefore, 53.9 kJ are produced.}$$

Chapter 11 Part 4: Titration

38.

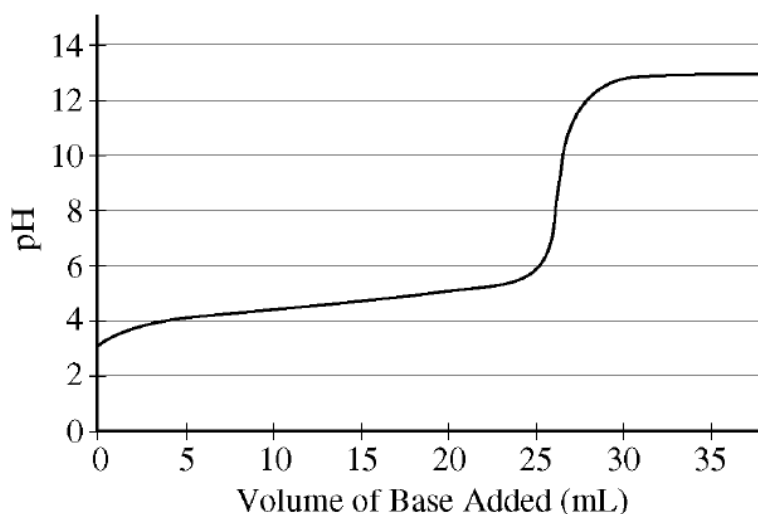


The graph above shows the titration curve that resulted when a sample of 0.1 *M* monoprotic acid was titrated with a solution of NaOH. Based on the graph, the pK_a of the acid is closest to

- (A) 3.0
- (B) 4.0
- (C) 6.0
- (D) 8.0
- (E) 12.0



39.



A student performs an acid-base titration and plots the experimental results in the graph above. Which of the following statements best explains the experimental findings?

Chapter 11 Part 4: Titration

- (A) A strong acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} = 7$.
- (B) A strong acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} > 7$.
- (C) A weak acid was titrated with a strong base, as evidenced by the equivalence point at $\text{pH} > 7$. ✓
- (D) A weak acid was titrated with a weak base, as evidenced by the equivalence point at pH approximately 7.
-

Refer to the following.

Concentration (M)	pH of Acid 1	pH of Acid 2	pH of Acid 3	pH of Acid 4
0.010	3.44	2.00	2.92	2.20
0.050	3.09	1.30	2.58	1.73
0.10	2.94	1.00	2.42	1.55
0.50	2.69	0.30	2.08	1.16
1.00	2.44	0.00	1.92	0.98

The pH of solutions of four acids prepared at various concentrations were measured and recorded in the table above.

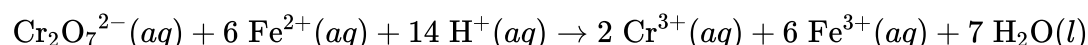
The four acids are, in no particular order, chlorous, hydrochloric, lactic, and propanoic.

40. If equal volumes of the four acids at a concentration of $0.50\ M$ are each titrated with a strong base, which will require the greatest volume of base to reach the equivalence point?
- (A) Acid 1
- (B) Acid 2
- (C) Acid 3
- (D) All the acids will require the same volume of base to reach the equivalence point. ✓
-

Chapter 11 Part 4: Titration

41. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student determines the mass percent of iron in a sample of iron ore by doing a redox titration. The student finds the mass of the sample and then dissolves it in acid to a total volume of 25.00 mL. Using a buret, the student titrates the 25.00 mL solution with 0.017 M $\text{K}_2\text{Cr}_2\text{O}_7$. The following reaction occurs during the titration.



The student collects the following data.

Mass of iron ore in original sample	0.36 g
Buret reading at end point	45.52 mL
Initial buret reading	15.05 mL

- (a) How many moles of $\text{Fe}^{2+}(\text{aq})$ reacted with the $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$?
- (b) Assuming that all the iron in the ore was Fe^{2+} , calculate the mass percent of iron in the ore sample.
- (c) The student learns that the actual mass percent of iron in the ore sample is greater than the experimental value calculated in part (b). The student claims that the difference between the actual and experimental values could have been caused by rinsing the buret with distilled water just prior to filling it with 0.017 M $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$. Do you agree with the student's claim? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets **all** the following criteria.

- ☐ The response provides the correct calculated number of moles of $\text{Cr}_2\text{O}_7^{2-}$.
 $45.52 \text{ mL} - 15.05 \text{ mL} = 30.47 \text{ mL}$
 $0.03047 \text{ L} \times \frac{0.017 \text{ mol Cr}_2\text{O}_7^{2-}}{1 \text{ L}} = 5.2 \times 10^{-4} \text{ mol Cr}_2\text{O}_7^{2-}$
- ☐ The response provides the correct calculated number of moles of Fe^{2+} , consistent with the moles of $\text{Cr}_2\text{O}_7^{2-}$.

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$$5.2 \times 10^{-4} \text{ mol Cr}_2\text{O}_7^{2-} \times \frac{6 \text{ mol Fe}^{2+}}{1 \text{ mol Cr}_2\text{O}_7^{2-}} = 0.0031 \text{ mol Fe}^{2+}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value, consistent with the moles of Fe^{2+} .

$$0.0031 \text{ mol Fe}^{2+} \times \frac{55.85 \text{ g}}{1 \text{ mol}} = 0.17 \text{ g Fe}$$

$$\frac{0.17 \text{ g Fe}}{0.36 \text{ g sample}} \times 100 = 47\%$$

(48% without rounding in earlier steps)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



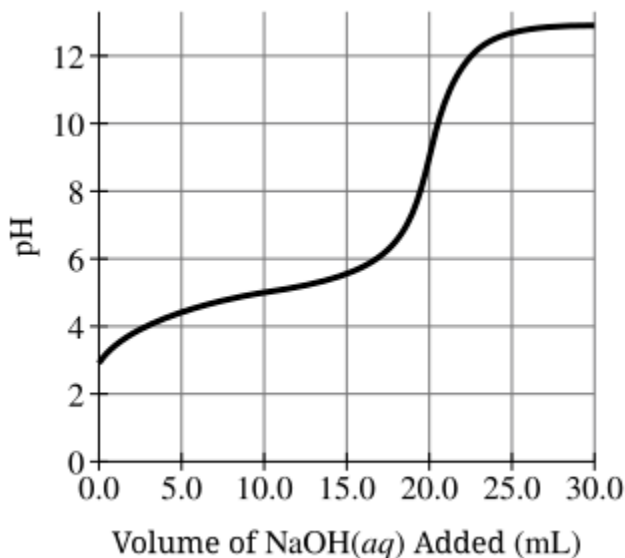
0	1
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The response provides the correct answer and a valid justification.

Disagree. The distilled water dilutes the $\text{K}_2\text{Cr}_2\text{O}_7(aq)$ solution, requiring a greater volume to reach the endpoint of the titration. This increased volume, when multiplied by 0.017 M (the concentration of $\text{K}_2\text{Cr}_2\text{O}_7(aq)$), would result in a greater calculated number of moles of iron (and thus a higher experimental % of iron).

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42.



A 0.10 M solution of a weak acid is titrated with 0.10 M $\text{NaOH}(aq)$. Based on the titration curve above, which of the following could be the identity of the weak acid?

- (A) HCN with a K_a value of 5×10^{-10}
- (B) HOCl with a K_a value of 3×10^{-8}
- (C) $\text{CH}_3\text{CH}_2\text{COOH}$ with a K_a value of 1×10^{-5} ✓
- (D) ClCH_2COOH with a K_a value of 1×10^{-3}

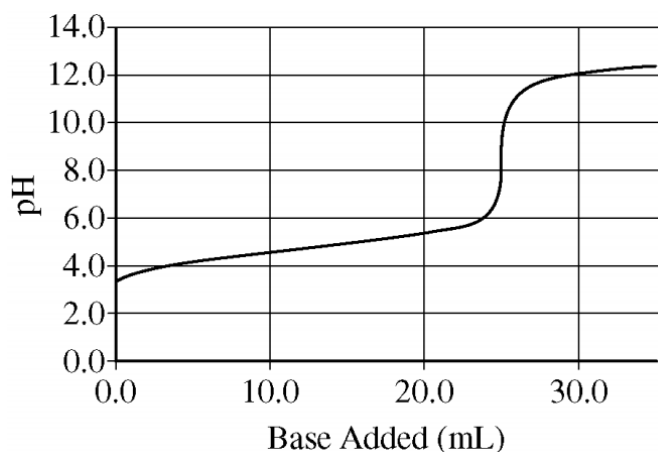
43. When a buret is rinsed before a titration, which of the techniques below is the best procedure?

- (A) Rinse the buret one time with some of the titrant solution.
- (B) Rinse the buret one time with some of the titrant solution and then dry the buret in an oven.
- (C) Rinse the buret two times: once with some of the titrant solution, then once with distilled water.
- (D) Rinse the buret two times: each time with some of the titrant solution. ✓
- (E) Rinse the buret two times: each time with distilled water.

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A 0.35 g sample of $\text{Li}(s)$ is placed in an Erlenmeyer flask containing 100 mL of water at 25°C . A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the $\text{Li}(s)$ has reacted with $\text{H}_2\text{O}(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below



44. On the basis of the pH curve, the pK_a value of the acid is closest to

(A) 4

(B) 5

(C) 8

(D) 12

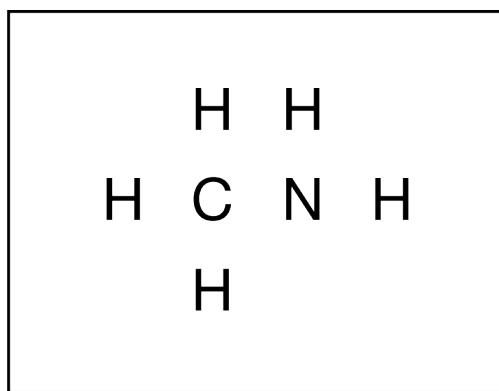


Chapter 11 Part 4: Titration

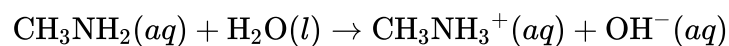
45. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Methylamine is a weak base with the formula CH_3NH_2 .

(a) In the following box, complete the Lewis electron-dot diagram for a molecule of methylamine. Show all bonding and nonbonding valence electrons.



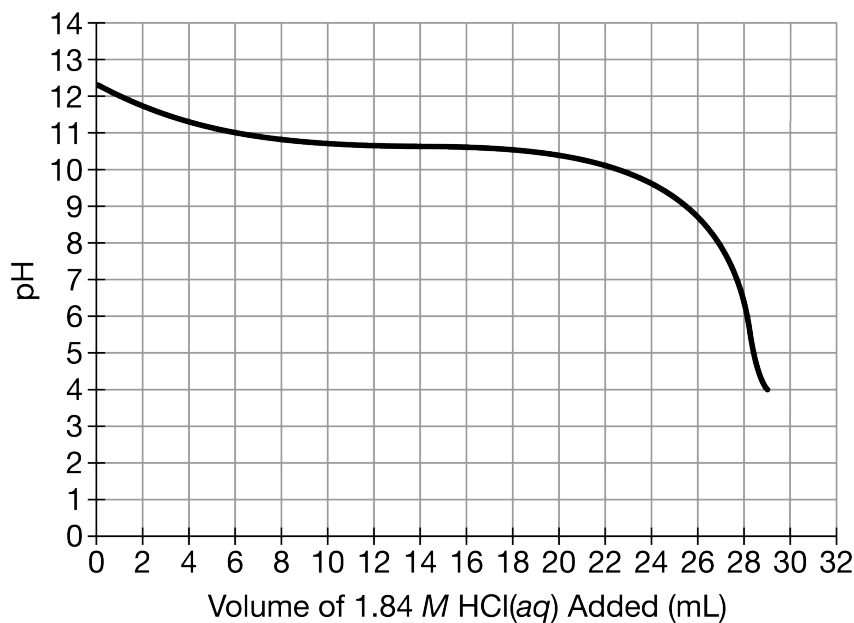
The following chemical equation represents the reaction that occurs when methylamine dissolves in water to form a basic solution.



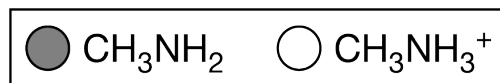
(b) The pH of 2.65 M $\text{CH}_3\text{NH}_2(aq)$ is 12.54. Determine the value of K_b for methylamine.

A 25.00 mL sample of a $\text{CH}_3\text{NH}_2(aq)$ solution of unknown concentration is titrated with 1.84 M $\text{HCl}(aq)$. Following is a graph that shows pH versus the volume of 1.84 M $\text{HCl}(aq)$ added during the titration.

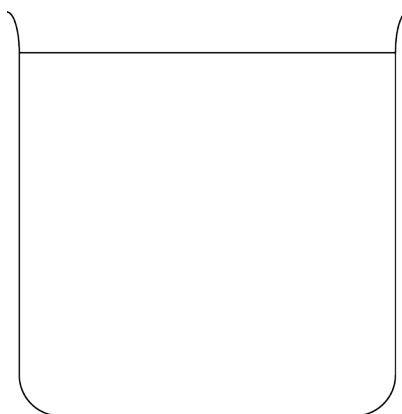
Chapter 11 Part 4: Titration



(c) If 28.25 mL of 1.84 M HCl(aq) was required to reach the equivalence point, calculate the concentration of the CH₃NH₂(aq) solution of unknown concentration.



(d) Using the symbols in the legend above, draw particles in the following beaker to represent the relative amounts of the two species, CH₃NH₂ and CH₃NH₃⁺, in the solution after the first 5.00 mL of titrant had been added to the CH₃NH₂(aq) solution.



Beaker after 5.00 mL of
1.84 M HCl(aq) Added

(e) Explain why the titration curve shows only a small change in pH per volume of acid added when the total amount of acid added is about 14.0 mL. Include a balanced chemical equation as part of your answer.

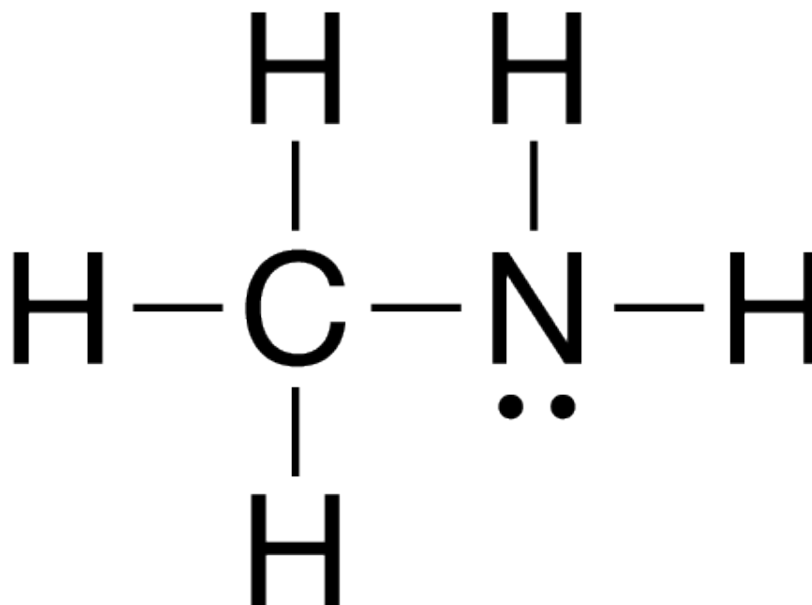
Chapter 11 Part 4: Titration**Part (a)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student's drawing includes exactly 14 electrons; one pair of electrons between every two adjacent atoms plus one lone pair of electrons on the nitrogen atom. Electron pairs can be represented either by dots, or line segments. See example drawing below.

**Part (b)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The points should be treated independently in that the correct use of incorrect prior results can earn credit.

Chapter 11 Part 4: Titration



0	1	2	3
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The response meets all of the following three criteria:

- ☐ The correct calculation of $[\text{OH}^-]$:
 $\text{pH} = 12.54 \Rightarrow \text{pOH} = (14.00 - 12.54) = 1.46 \Rightarrow [\text{OH}^-] = 10^{-1.46} = 3.47 \times 10^{-2} \text{ M}$
- ☐ The use of the correct expression for K_b :
 As methylamine reacts with water, $[\text{CH}_3\text{NH}_3^+] = [\text{OH}^-]$. Let x represent $[\text{OH}^-]$, then
 $K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2]} = \frac{x^2}{2.65 - x} \approx \frac{x^2}{2.65}$ because $x \ll 2.65$.
- ☐ The correct substitution of the numbers and calculation of the value:
 Substituting the value of $[\text{OH}^-]$ for x , $K_b = \frac{(3.47 \times 10^{-2})^2}{2.65} = 4.54 \times 10^{-4}$

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The points should be treated independently in that the correct use of incorrect prior results can earn credit.



0	1	2
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The response includes both of the following criteria:

- ☐ The correct calculation of moles of $\text{HCl}(aq)$ added:
 $0.02825 \text{ L HCl}(aq) \text{ solution} \times \frac{1.84 \text{ mol HCl}(aq)}{1.000 \text{ L solution}} = 0.05198 \text{ mol HCl}(aq)$
- ☐ The correct calculation of the concentration of the $\text{CH}_3\text{NH}_2(aq)$ solution:
 $0.05198 \text{ mol HCl}(aq) \text{ for equivalence} \Rightarrow 0.05198 \text{ mol CH}_3\text{NH}_2 \text{ present in the sample, thus}$
 $[\text{CH}_3\text{NH}_2] = \frac{0.05198 \text{ mol CH}_3\text{NH}_2}{0.02500 \text{ L}} = 2.08 \text{ M}$.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows more CH_3NH_2 particles than CH_3NH_3^+ particles.

Part (e)

Chapter 11 Part 4: Titration

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

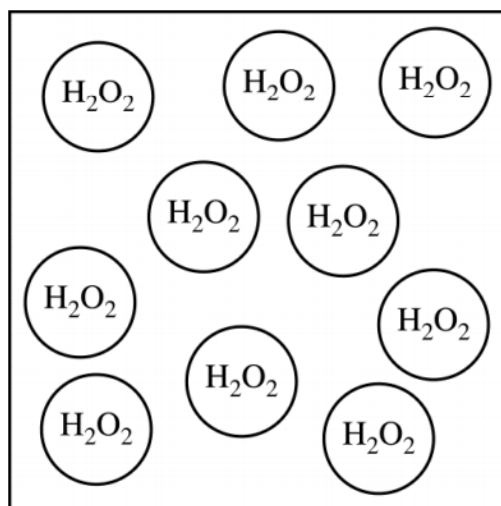


0	1	2
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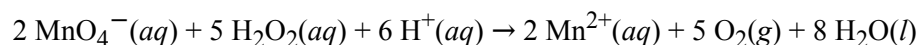
The response includes both of the following criteria:

- ☐ An explanation that, near the half equivalence point, the concentrations of the base and of its conjugate acid are approximately equal (the buffer zone). The effect on pH of adding more acid to this system is partially counteracted because the added acid reacts with the base in the solution.
- ☐ The balanced equation $\text{H}_3\text{O}^+ + \text{CH}_3\text{NH}_2 \rightarrow \text{H}_2\text{O} + \text{CH}_3\text{NH}_3^+$

46.



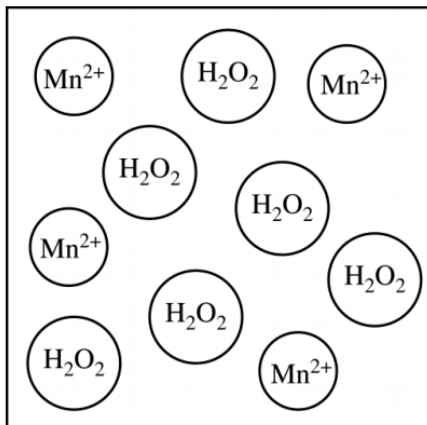
A particle view of a sample of $\text{H}_2\text{O}_2(aq)$ is shown above. The $\text{H}_2\text{O}_2(aq)$ is titrated with $\text{KMnO}_4(aq)$, as represented by the equation below.



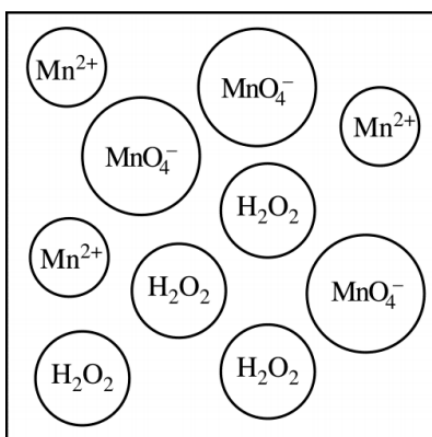
Which of the following particle views best represents the mixture when the titration is halfway to the equivalence point? (H_2O molecules and H^+ ions are not shown.)

Chapter 11 Part 4: Titration

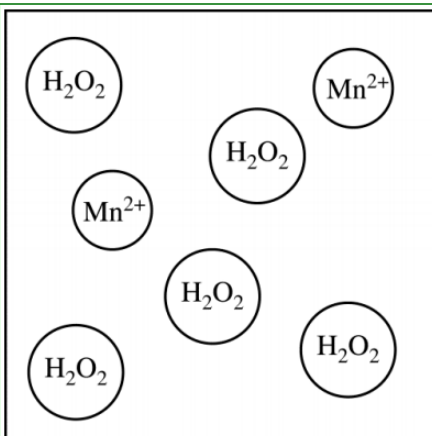
(A)



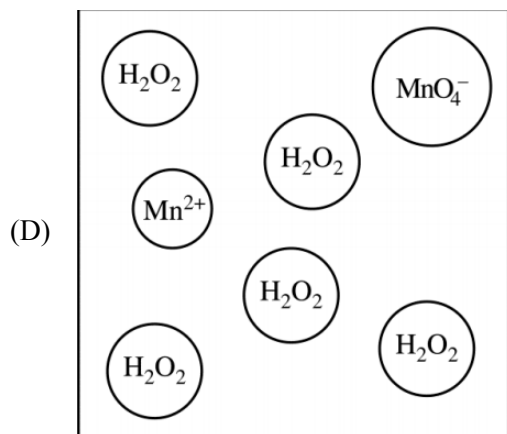
(B)



(C)



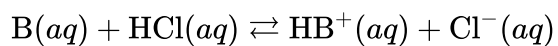
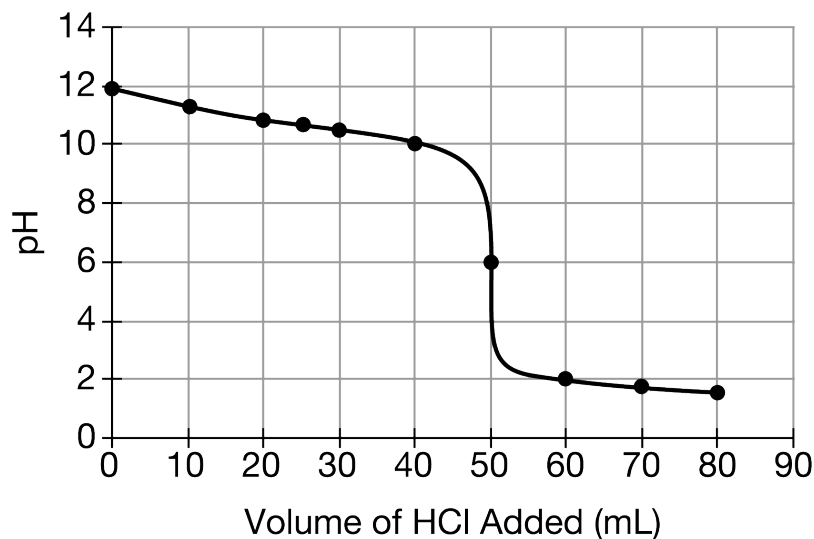
Chapter 11 Part 4: Titration



47. A 0.08 *M* solution of CH_3COOH ($\text{p}K_a = 4.74$) is titrated with 0.10 *M* $\text{NaOH}(aq)$. What is the pH at the equivalence point of the titration and why?
- (A) $\text{pH} < 7$, because $\text{NaOH}(aq)$ is a strong base.
- (B) $\text{pH} = 7$, because the titration reaction is a neutralization reaction.
- (C) $\text{pH} > 7$, because $\text{CH}_3\text{COO}^-(aq)$ is a weak base. ✓
- (D) $\text{pH} > 7$, because the concentration of $\text{NaOH}(aq)$ is greater than that of $\text{CH}_3\text{COOH}(aq)$.
48. When a strong acid is titrated with a strong base using phenolphthalein as an indicator, the color changes abruptly at the endpoint of the titration and can be switched back and forth by the addition of only one drop of acid or base. The reason for the abruptness of this color change is that
- (A) a large change in pH occurs near the endpoint of the titration ✓
- (B) a buffer solution exists at the endpoint of the titration
- (C) phenolphthalein is a strong proton donor
- (D) the pH of water is very resistant to change
- (E) phenolphthalein is much more sensitive to the pH of a solution than most other indicators

Chapter 11 Part 4: Titration

49.



The graph above shows the titration curve for an aqueous solution of a weak base, B, with HCl as the titrant. Based on the graph, which of the following best estimates the $\text{p}K_a$ of HB^+ ?

(A) 12.0

(B) 10.8

(C) 6.0

(D) 1.8

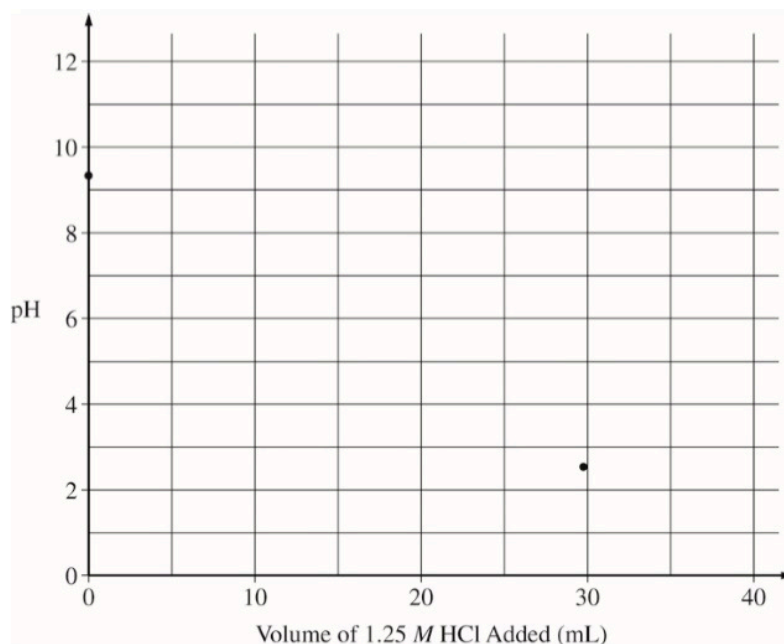


Chapter 11 Part 4: Titration

50. Potassium sorbate, $\text{KC}_6\text{H}_7\text{O}_2$ (molar mass 150. g/mol) is commonly added to diet soft drinks as a preservative. A stock solution of $\text{KC}_6\text{H}_7\text{O}_2(\text{aq})$ of known concentration must be prepared. A student titrates 45.00 mL of the stock solution with 1.25 M $\text{HCl}(\text{aq})$ using both an indicator and a pH meter. The value of K_a for sorbic acid, $\text{C}_6\text{H}_7\text{O}_2$, is 1.7×10^{-5} .
- Write the net-ionic equation for the reaction between $\text{KC}_6\text{H}_7\text{O}_2(\text{aq})$ and $\text{HCl}(\text{aq})$.
 - A total of 29.95 mL of 1.25 M $\text{HCl}(\text{aq})$ is required to reach the equivalence point. Calculate $[\text{KC}_6\text{H}_7\text{O}_2]$ in the stock solution.
 - The pH at the equivalence point of the titration is measured to be 2.54. Which of the following indicators would be the best choice for determining the end point of the titration? Justify your answer.

Indicator	$\text{p}K_a$
Phenolphthalein	9.3
Bromothymol blue	7.0
Methyl red	5.0
Thymol blue	2.0
Methyl violet	0.80

- Calculate the pH at the half-equivalence point.
- The initial pH and the equivalence point are plotted on the graph below. Accurately sketch the titration curve on the graph below. Mark the position of the half-equivalence point on the curve with an X.

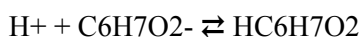


Chapter 11 Part 4: Titration

- f. The pH of the soft drink is 3.37 after the addition of the $\text{KC}_6\text{H}_7\text{O}_2(aq)$. Which species, $\text{HC}_6\text{H}_7\text{O}_2$ or $\text{C}_6\text{H}_7\text{O}_2^-$, has a higher concentration in the soft drink? Justify your answer.

Part A

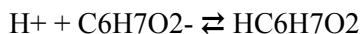
1 point is earned for the net-ionic equation.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the net-ionic equation.



Part B

1 point is earned for the moles of HCl at the equivalence point.

$$\frac{1.25 \text{ mol HCl}}{1000 \text{ mL}} = \frac{x \text{ mol HCl}}{29.95 \text{ mL}} \quad x = 0.0374 \text{ mol HCl}$$

1 point is earned for the correct answer.

$$\frac{0.0374 \text{ mol C}_6\text{H}_7\text{O}_2^{\text{msup}}}{45.0 \text{ mL}} = \frac{x \text{ mol C}_6\text{H}_7\text{O}_2^{\text{msup}}}{1000 \text{ mL}} \Rightarrow 0.832M$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the moles of HCl at the equivalence point.

$$\frac{1.25 \text{ mol HCl}}{1000 \text{ mL}} = \frac{x \text{ mol HCl}}{29.95 \text{ mL}} \quad x = 0.0374 \text{ mol HCl}$$

1 point is earned for the correct answer.

$$\frac{0.0374 \text{ mol C}_6\text{H}_7\text{O}_2^-}{45.0 \text{ mL}} = \frac{x \text{ mol C}_6\text{H}_7\text{O}_2^-}{1000 \text{ mL}} \Rightarrow 0.832M$$

Part C

1 point is earned for the correct indicator.

Chapter 11 Part 4: Titration

1 point is earned for the correct justification.

Thymol blue; it has a pK_a close to the pH at the equivalence point, so it will change color near the equivalence point.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct indicator.

1 point is earned for the correct justification.

Thymol blue; it has a pK_a close to the pH at the equivalence point, so it will change color near the equivalence point.

Part D

1 point is earned for the correct pH.

$$\text{pH} = \text{p}K_a = -\log (1.7 \times 10^{-5}) = 4.77$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct pH.

$$\text{pH} = \text{p}K_a = -\log (1.7 \times 10^{-5}) = 4.77$$

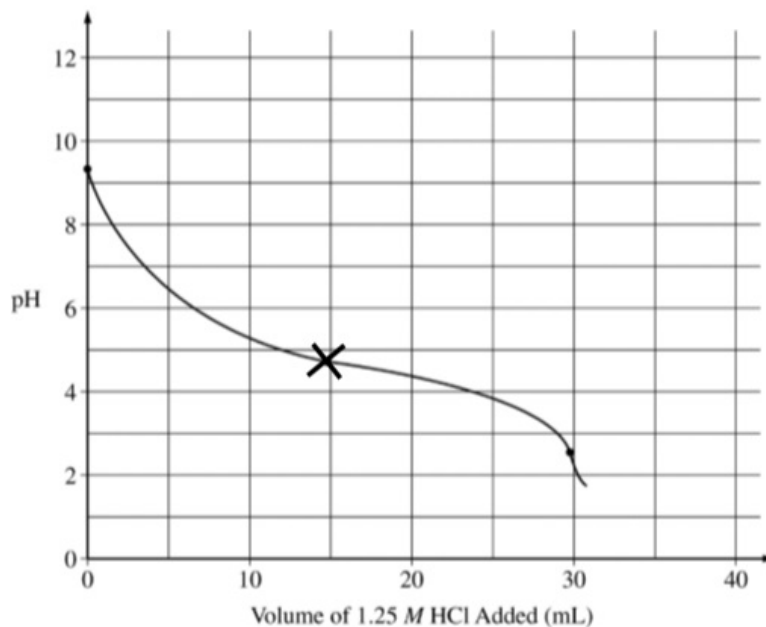
Part E

1 point is earned for a half-equivalence point consistent with the answer to part (d) and at the correct volume.

1 point is earned for a curve that levels off to a relatively horizontal slope through the half-equivalence point.

1 point is earned for a relatively steep negative slope through the equivalence point.

Chapter 11 Part 4: Titration



[The pH curve should have the correct shape.]



0	1	2	3
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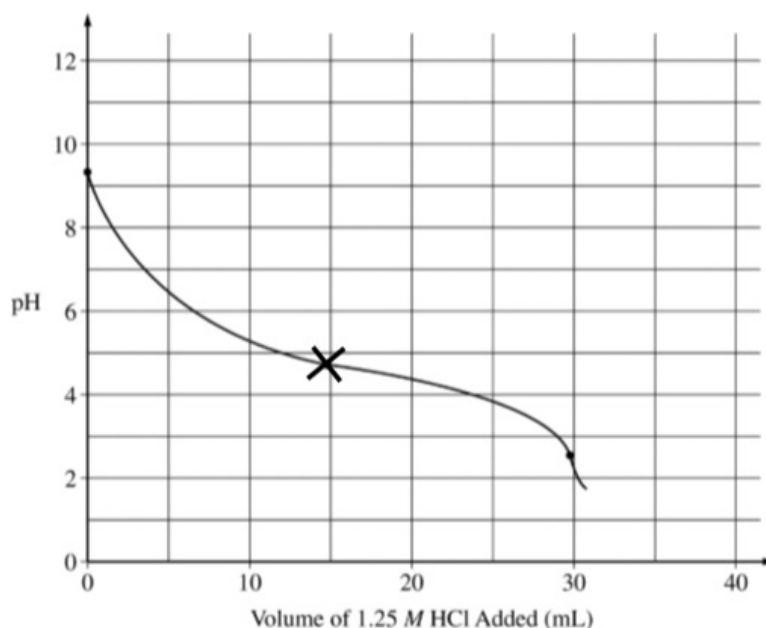
The student response earns three of the following points:

1 point is earned for a half-equivalence point consistent with the answer to part (d) and at the correct volume.

1 point is earned for a curve that levels off to a relatively horizontal slope through the half-equivalence point.

1 point is earned for a relatively steep negative slope through the equivalence point.

Chapter 11 Part 4: Titration



[The pH curve should have the correct shape.]

Part F

1 point is earned for identifying the correct species and for making a comparison involving the pH (with or without calculation).

$$\text{For sorbic acid, } K_a = \frac{[\text{H}^+][\text{C}_6\text{H}_7\text{O}_2^{\text{msup}}]}{[\text{HC}_6\text{H}_7\text{O}_2]},$$

$$\text{thus } \frac{[\text{C}_6\text{H}_7\text{O}_2^{\text{msup}}]}{[\text{HC}_6\text{H}_7\text{O}_2]} = \frac{K_a}{[\text{H}^+]} = \frac{1.7 \times 10^{-5}}{10^{-3.37}} \approx 0.04$$

$$\Rightarrow [\text{HC}_6\text{H}_7\text{O}_2] > [\text{C}_6\text{H}_7\text{O}_2^-]$$

OR

The concentrations of $\text{HC}_6\text{H}_7\text{O}_2$ and $\text{C}_6\text{H}_7\text{O}_2^-$ are equal at the half-equivalence point. A pH of 3.37 is lower than that at the half-equivalence point, so the protonated form, $\text{HC}_6\text{H}_7\text{O}_2$, has a higher concentration in the soft drink.



0

1

The student response earns one of the following points:

1 point is earned for identifying the correct species and for making a comparison involving the pH (with or without calculation).

$$\text{For sorbic acid, } K_a = \frac{[\text{H}^+][\text{C}_6\text{H}_7\text{O}_2^-]}{[\text{HC}_6\text{H}_7\text{O}_2]},$$

Chapter 11 Part 4: Titration

$$\text{thus } \frac{[\text{C}_6\text{H}_7\text{O}_2^-]}{[\text{HC}_6\text{H}_7\text{O}_2]} = \frac{K_a}{[\text{H}^+]} = \frac{1.7 \times 10^{-5}}{10^{-3.37}} \approx 0.04$$

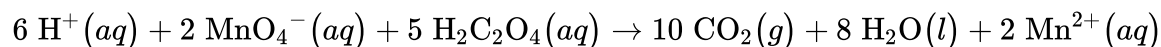
$$\Rightarrow [\text{HC}_6\text{H}_7\text{O}_2] > [\text{C}_6\text{H}_7\text{O}_2^-]$$

OR

The concentrations of $\text{HC}_6\text{H}_7\text{O}_2$ and $\text{C}_6\text{H}_7\text{O}_2^-$ are equal at the half-equivalence point. A pH of 3.37 is lower than that at the half-equivalence point, so the protonated form, $\text{HC}_6\text{H}_7\text{O}_2$, has a higher concentration in the soft drink.

Chapter 11 Part 4: Titration

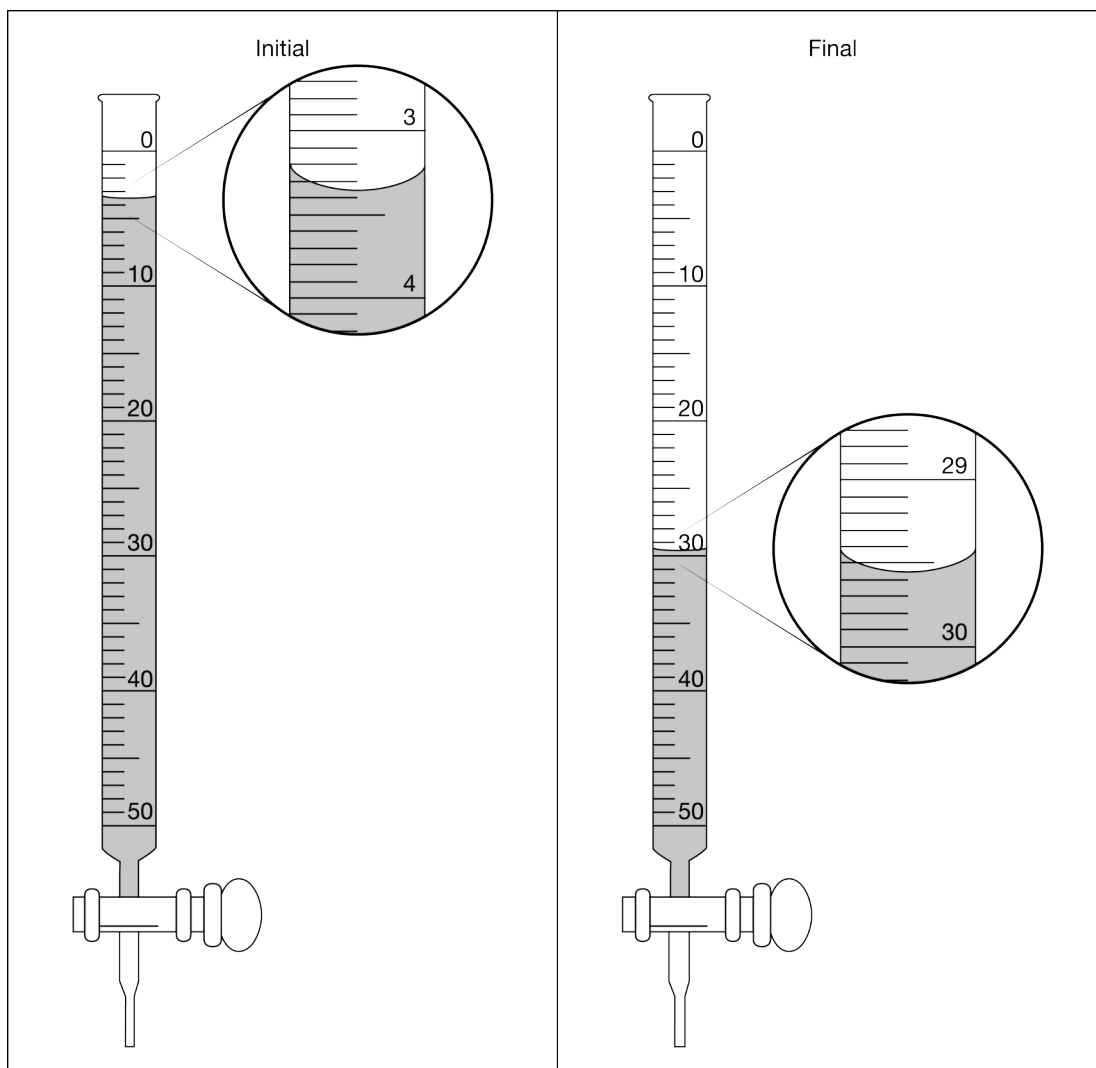
51. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A student dissolved a 0.139 g sample of oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$, in water in an Erlenmeyer flask. Then the student titrated the $\text{H}_2\text{C}_2\text{O}_4$ solution in the flask with a solution of KMnO_4 , which has a dark purple color. The balanced chemical equation for the reaction that occurred during the titration is shown above.

- (a) Identify the species that was reduced in the titration reaction. Justify your answer in terms of oxidation numbers.
- (b) The student used a 50.0 mL buret to add the $\text{KMnO}_4(aq)$ to the $\text{H}_2\text{C}_2\text{O}_4(aq)$ until a faint lavender color was observed in the flask, an indication that the end point of the titration had been reached. The initial and final volume readings of the solution in the buret are shown below. Write down the initial reading and the final reading and use them to determine the volume of $\text{KMnO}_4(aq)$ that was added during the titration.

Chapter 11 Part 4: Titration



(c) Given that the concentration of $\text{KMnO}_4(aq)$ was 0.0235 M , calculate the number of moles of MnO_4^- ions that completely reacted with the $\text{H}_2\text{C}_2\text{O}_4$.

(d) The student proposes to perform another titration using a 0.139 g sample of $\text{H}_2\text{C}_2\text{O}_4$, but this time using $0.00143\text{ M KMnO}_4(aq)$ in the buret. Would this titrant concentration be a reasonable choice to use if the student followed the same procedure and used the same equipment as before? Justify your response.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

Chapter 11 Part 4: Titration

The response indicates that MnO_4^- is reduced to Mn^{2+} as the oxidation number of Mn changes from +7 to +2, a gain of 5 electrons.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows a calculation similar to the following, consistent with values measured from the bottom of the meniscus on the magnified glassware illustration.

$$29.55 \text{ mL} - 3.35 \text{ mL} = 26.20 \text{ mL}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows a calculation similar to the following and is consistent with the answer in part (b) and maintains 3 significant figures from the molarity.

$$(0.02620 \text{ L})(0.0235 \text{ mol/L}) = 0.000616 \text{ mol}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

- ☐ · The response indicates “No”.
- ☐ · The response indicates that the 0.00143 *M* titrant solution is so diluted that the volume of titrant needed to reach the end point would be much greater than the 50 mL capacity of the buret.

Chapter 11 Part 4: Titration

52. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student reacts 0.300 g of methyl salicylate ($\text{C}_8\text{H}_8\text{O}_3$) with a stoichiometric amount of a strong base. This product is then acidified to produce salicylic acid crystals ($\text{HC}_7\text{H}_5\text{O}_3$).

(a) For every 1 mole of $\text{C}_8\text{H}_8\text{O}_3$ (molar mass 152.15 g/mol) reactant used, 1 mole of salicylic acid crystals ($\text{HC}_7\text{H}_5\text{O}_3$, molar mass 138.12 g/mol) is produced. Calculate the maximum mass, in grams, of $\text{HC}_7\text{H}_5\text{O}_3$ that could be produced in this reaction.

As part of the experimental procedure to purify the $\text{HC}_7\text{H}_5\text{O}_3$ crystals after the reaction is complete, the crystals are filtered from the reaction mixture, rinsed with distilled water, and dried. Some physical properties of $\text{HC}_7\text{H}_5\text{O}_3$ are given in the following table.

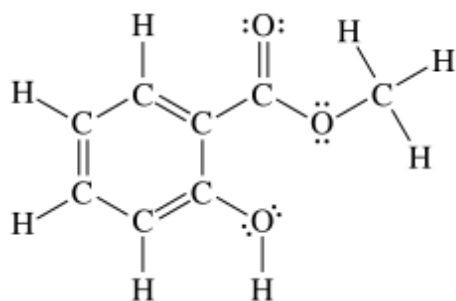
Properties of Salicylic Acid ($\text{HC}_7\text{H}_5\text{O}_3$)	
Melting point	159 °C
Solubility in H_2O at 25 °C	2.2 g/L
Specific heat capacity	1.17 J/(g·°C)
Heat of fusion	27.1 kJ/mol

(b) The student's experiment results in an 87% yield of dry $\text{HC}_7\text{H}_5\text{O}_3$. The student suggests that some of the $\text{HC}_7\text{H}_5\text{O}_3$ crystals dissolved in the distilled water during the rinsing step. Is the student's claim consistent with the calculated percent yield value? Justify your answer.

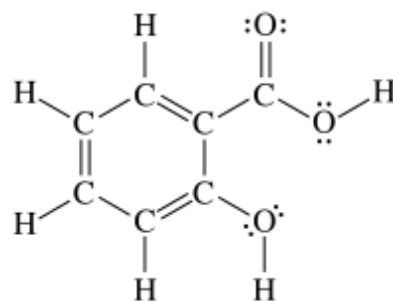
(c) Given the physical properties in the table, calculate the quantity of heat that must be absorbed to increase the temperature of a 0.105 g sample of dry $\text{HC}_7\text{H}_5\text{O}_3$ (molar mass 138.12 g/mol) crystals from 25 °C to the melting point of 159 °C and melt the crystals completely.

The structures and melting points for methyl salicylate and salicylic acid are shown.

Chapter 11 Part 4: Titration



Methyl Salicylate
Melting Point: -9°C

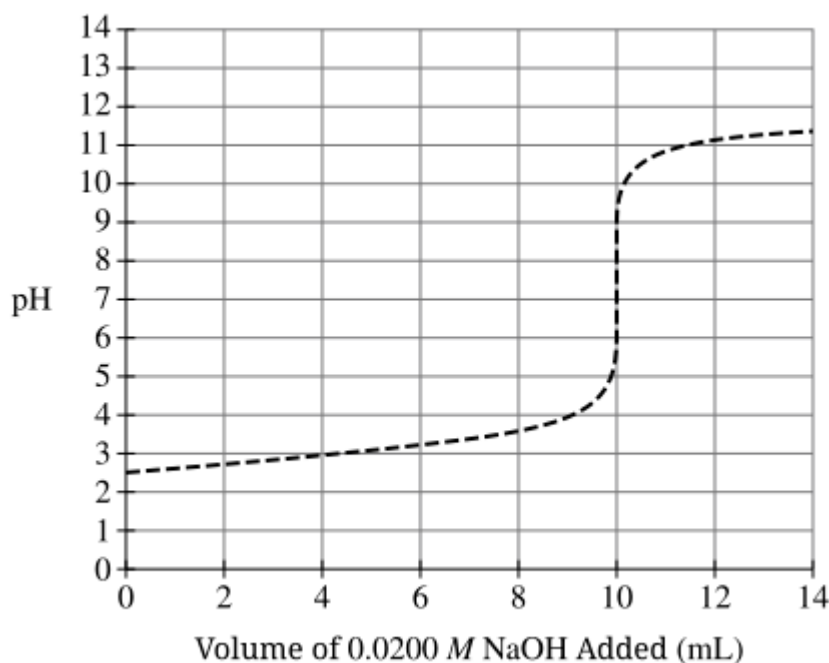


Salicylic Acid
Melting Point: 159°C

(d) The same three types of intermolecular forces (London dispersion forces, dipole-dipole interactions, and hydrogen bonding) exist among molecules of each substance. Explain why the melting point of salicylic acid is higher than that of methyl salicylate.

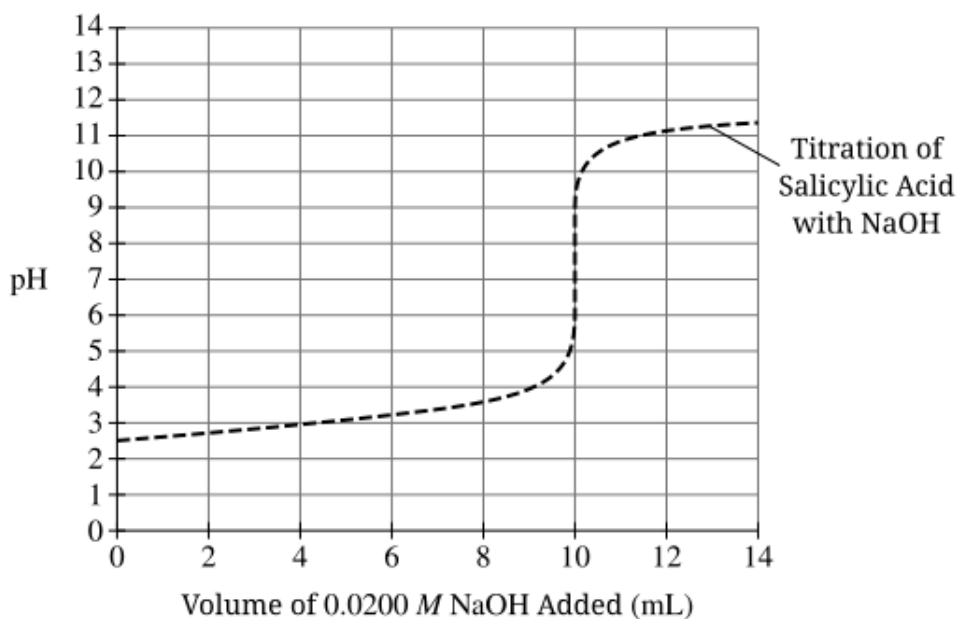
The student titrates 20.0 mL of 0.0100M $\text{HC}_7\text{H}_5\text{O}_3(aq)$ with 0.0200M NaOH, using a probe to monitor the pH of the solution. The data are plotted producing the following titration curve.

Chapter 11 Part 4: Titration



- (e) Using the information in the graph, estimate the pK_a of $HC_7H_5O_3$. _____
- (f) When the pH of the titration mixture is 4.00, is there a higher concentration of the weak acid, $HC_7H_5O_3$, or its conjugate base, $C_7H_5O_3^-$, in the flask? Justify your answer.
- (g) The student researches benzoic acid ($HC_7H_5O_2$) and finds that it has similar properties to salicylic acid ($HC_7H_5O_3$). The K_a for benzoic acid is 6.3×10^{-5} . Calculate the value of pK_a for benzoic acid.
- (h) The student performs a second titration, this time titrating 20.0 mL of a 0.0100 M benzoic acid solution with 0.0200 M NaOH. Sketch the curve that would result from this titration of benzoic acid on the following graph, which already shows the original curve from the titration of 20.0 mL of 0.0100 M salicylic acid. The initial pH of the benzoic acid solution is 3.11.

Chapter 11 Part 4: Titration



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct calculated value.

$$\begin{aligned}
 & \bullet \ 0.300 \text{ g C}_8\text{H}_8\text{O}_3 \times \frac{1 \text{ mol C}_8\text{H}_8\text{O}_3}{152.15 \text{ g}} \times \frac{1 \text{ mol HC}_7\text{H}_5\text{O}_3}{1 \text{ mol C}_8\text{H}_8\text{O}_3} \times \frac{138.12 \text{ g}}{1 \text{ mol HC}_7\text{H}_5\text{O}_3} \\
 & = 0.272 \text{ g HC}_7\text{H}_5\text{O}_3
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct answer and a valid justification.

- Yes (consistent). Because the acid is soluble in water, some crystals may dissolve during rinsing, causing the mass of the collected precipitate to be lower than expected. This would lead to a percent yield less than 100%.

Chapter 11 Part 4: Titration

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets **all** the following criteria.

- ☐ The response provides the correct calculated value for the heat absorbed during the temperature change.

$$\circ q_{\text{heat}} = mc\Delta T = (0.105 \text{ g})(1.17 \text{ J/(g} \cdot ^\circ\text{C)})(159^\circ\text{C} - 25^\circ\text{C}) = 16.5 \text{ J}$$

- ☐ The response provides the correct calculated value for the heat absorbed during the phase change.

$$\circ q_{\text{melt}} = 0.105 \text{ g} \times \frac{1 \text{ mol}}{138.12 \text{ g}} \times \frac{27,100 \text{ J}}{1 \text{ mol}} = 20.6 \text{ J}$$

- ☐ The response provides the correct calculated value for the total heat absorbed.

$$\circ q_{\text{total}} = q_{\text{heat}} + q_{\text{melt}} = 16.5 \text{ J} + 20.6 \text{ J} = 37.1 \text{ J}$$

- ☐ The response provides the correct calculated value for the total heat absorbed based on the correct calculated values of the heat absorbed during both the temperature change and the phase change.

$$\circ q_{\text{total}} = q_{\text{heat}} + q_{\text{melt}} = 16.5 \text{ J} + 20.6 \text{ J} = 37.1 \text{ J}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

- Molecules of salicylic acid have more hydrogen bonding sites than molecules of methyl salicylate have, which leads to stronger intermolecular forces and a higher melting point for salicylic acid.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 4: Titration



0	1
---	---

The response provides the correct answer.

- The $\text{p}K_a$ is approximately 3.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer and a valid justification, consistent with part (e).

- The conjugate base, $\text{C}_7\text{H}_5\text{O}_3^-$. When $\text{pH} = 4$, the titration is beyond the half-equivalence point, where $[\text{HC}_7\text{H}_5\text{O}_3] = [\text{C}_7\text{H}_5\text{O}_3^-]$. Thus, the molar concentration of $[\text{C}_7\text{H}_5\text{O}_3^-]$ must be greater than $[\text{HC}_7\text{H}_5\text{O}_3]$

OR

- The conjugate base, $\text{C}_7\text{H}_5\text{O}_3^-$. Because the pH of the solution is greater than the $\text{p}K_a$ of the acid, the majority of the molecules will be deprotonated.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct calculated value.

- $\text{p}K_a = -\log(6.3 \times 10^{-5}) = 4.20$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

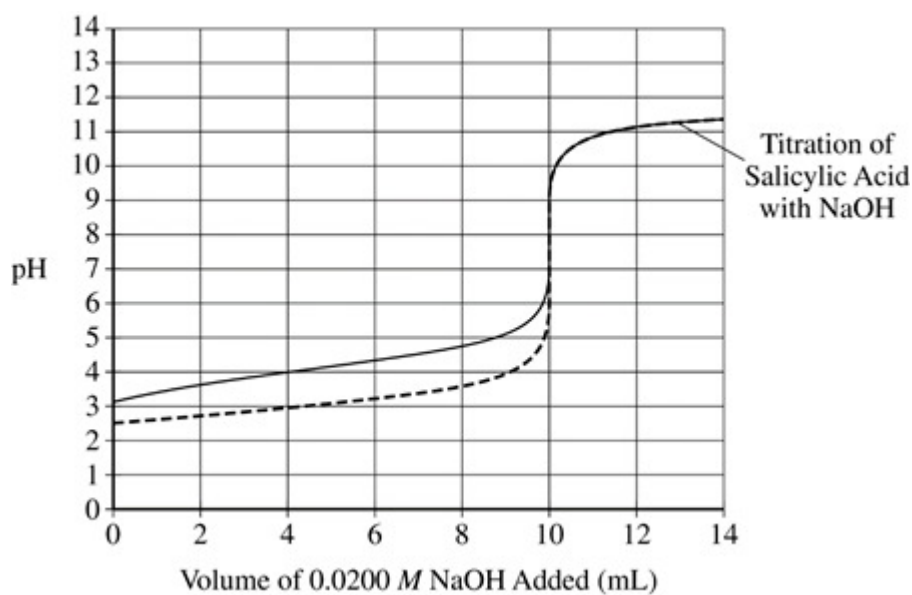
Chapter 11 Part 4: Titration



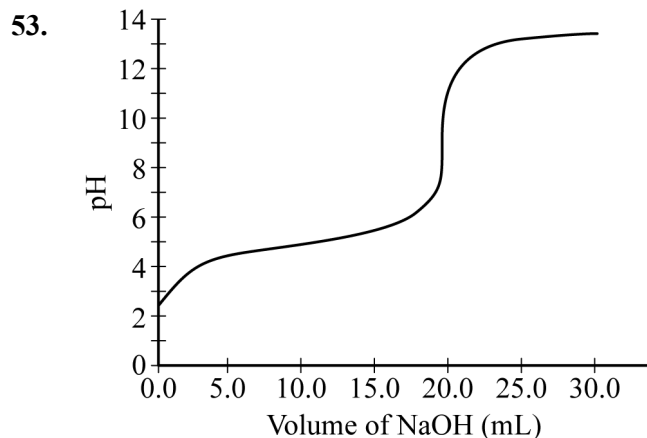
0	1	2
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The response meets **both** the following criteria.

- ☐ The curve shows a correct starting point and half-equivalence point, consistent with part (g).
 - The curve starts at $\text{pH} \approx 3.11$ and passes through the $\text{p}K_a$ calculated in part (g) at 5 milliliters. See example response below.
- ☐ The curve shows the correct equivalence point.
 - The curve inflects vertically at 10 mL, showing the same volume of base needed to reach the equivalence point.



Chapter 11 Part 4: Titration



Data collected during the titration of a 20.0 mL sample of a 0.10 M solution of a monoprotic acid with a solution of NaOH of unknown concentration are plotted in the graph above. Based on the data, which of the following are the approximate pK_a of the acid and the molar concentration of the NaOH?

(A)	pK_a	[NaOH]
	4.7	0.050 M

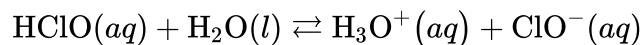
(B)	pK_a	[NaOH]
	4.7	0.10 M

(C)	pK_a	[NaOH]
	9.3	0.050 M

(D)	pK_a	[NaOH]
	9.3	0.10 M

Chapter 11 Part 4: Titration

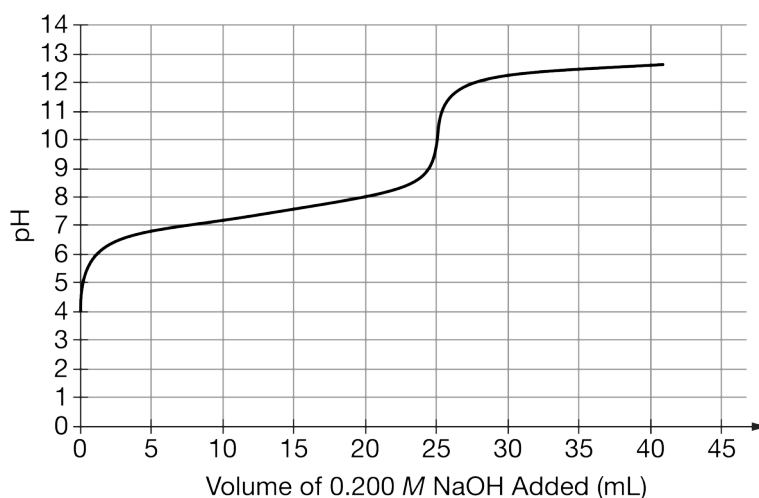
54. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The reaction between hypochlorous acid and water is represented above.

(a) Identify one of the conjugate acid-base pairs in the reaction.

(b) Shown in the graph below is the titration curve that results when 25.0 mL of 0.200 *M* HClO is titrated with 0.200 *M* NaOH. Carefully draw a second curve on the graph that would result from the titration of 25.0 mL of 0.200 *M* HClO₄ with 0.200 *M* NaOH.



(c) A student proposes creating a buffer by dissolving 0.010 mol of NaClO₄(*s*) in 100. mL of 0.100 *M* HClO₄. Explain why the resulting solution would not be a buffer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response gives one of the following pairs.

· HClO and ClO[−]

· H₂O and H₃O⁺

Chapter 11 Part 4: Titration

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

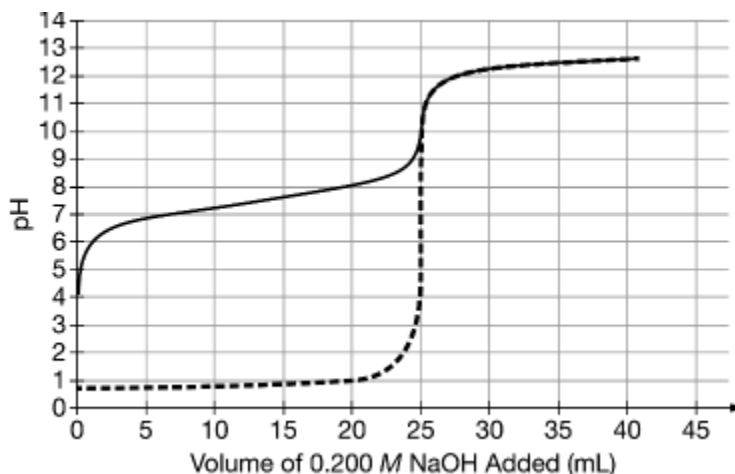


0	1	2
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The curve meets both of the following criteria.

- ☐ The curve has the shape of a titration curve, starts at a pH below the first curve, and ends above pH 7.
- ☐ The curve shows an equivalence point at 25 mL.

(See the following sample response.)



Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



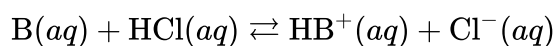
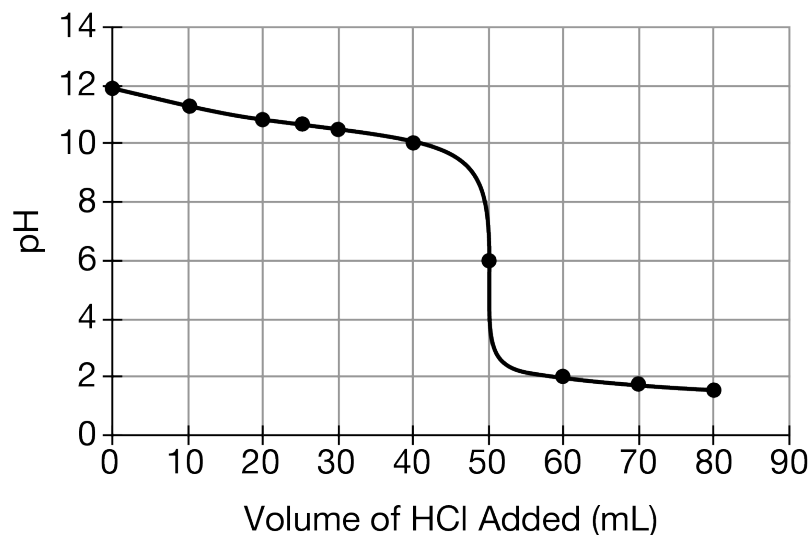
0	1
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The response gives an explanation such as the following.

A buffer solution contains a large concentration of both members in a weak acid/weak base conjugate pair. This solution would not be a buffer because HClO_4 is a strong acid and therefore ClO_4^- is negligibly basic and cannot react with added H_3O^+ .

Chapter 11 Part 4: Titration

55.



The reaction of a weak base, B, with HCl is represented by the equation above. The graph shows the titration curve for 25.0 mL of an aqueous solution of B titrated with 0.100 M HCl. Based on the graph, which of the following best estimates the initial concentration of the solution of the weak base?

(A) 0.20 M

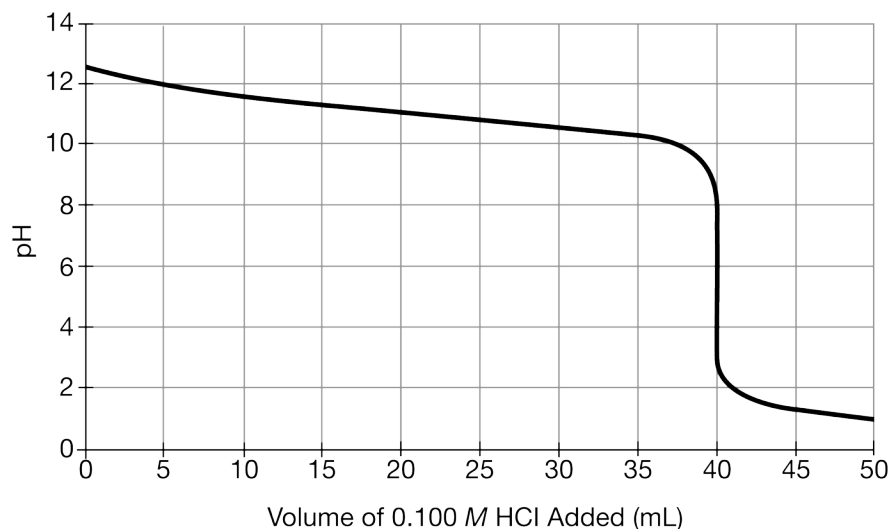
(B) 0.10 M

(C) 0.067 M

(D) 0.048 M



56.



A 40.0 mL sample of the weak base $\text{C}_5\text{H}_{11}\text{N}$ was titrated with 1.00 M HCl at 25 °C. Based on the resulting titration curve shown above, which of the following pairs provide the best estimates for the $\text{p}K_b$ and K_b of $\text{C}_5\text{H}_{11}\text{N}$?

Chapter 11 Part 4: Titration

- (A) $\text{p}K_b \approx 3.0$ and $K_b \approx 1 \times 10^{-3}$
- (B) $\text{p}K_b \approx 5.8$ and $K_b \approx 2 \times 10^{-6}$
- (C) $\text{p}K_b \approx 11.0$ and $K_b \approx 1 \times 10^{-11}$
- (D) $\text{p}K_b \approx 12.5$ and $K_b \approx 3 \times 10^{-2}$

